Castles in the Air

Extended chapter commentary

CHAPTER 6
The temple of thought

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CHAPTER 6 COMMENTARY

The temple of thought

We began this chapter's story with a philosophical brain-teaser, in the form of an argument that seems to establish the following remarkable fact:

Either there are Arithmetic statements
(i.e. Whole number glyph statements)
which we as humans will never be able to prove or disprove,
or there's something about our minds which
surpasses the power of any tube machine.

This famous conundrum is known as *Gödel's disjunction*, and we shall be discussing it in much more detail in Chapter 7. I don't want to say any more about it just yet — I included it here partly to give you something to ponder at this point, but also partly as a means of introducing a central question which, in various guises, will be occupying us for much of the rest of the book. The question is the one that Anna and Emily have now made the object of their inquiry: Can we nail down the concept of a 'valid mathematical proof' — or at least, that of a 'mathematically provable theorem' — via some definition that is itself mathematically precise? Our success in addressing an analogous question in the previous chapter — in giving a precise definition of a 'mechanically computable operation' — might well encourage us to expect a positive answer in this case too.

We've already laid a foundation for this investigation in Chapter 3, in that we now have a precisely defined *glyph language* within which mathematical statements can be expressed. We can therefore ask: what does it means to give a *proof* of a formal statement in this language? In other words, how can glyph formulae be manipulated? What are the rules for moving from one glyph statement to another that capture the process of mathematical proof?

It will be convenient to split our investigation of this into two halves. In this chapter, we are looking at the the purely 'logical' aspects of the language — at the rules for manipulating the six glyphs ¬, &, ∨, ⇒, ∀, ∃ — and seeing how logically valid steps of reasoning can be described purely as symbol-shunting operations. In the next chapter, we will turn our attention to the specifically 'mathematical' ingredients, such as the symbols 5, + and <.

What's the essential difference between the symbols I'm calling 'logical' and those I'm calling 'mathematical'? Ultimately, the distinction is more a matter of convenience than anything else, and I wouldn't want to press it too far. It's probably not fruitful to try to draw any hard-and-fast line between 'logic' and 'mathematics' — and there's also a point of view known as *logicism*, which we'll come to in Chapter 12, which says that the whole of mathematics should be seen as a *part* or outgrowth of logic. But for the time being, we'll go with the broad idea that 'logic' deals with the universal rules of reasoning that apply equally to *anything* we might wish to think about, whether Whole numbers, Real numbers, sets, meringues, cockatoos, widgets or jabberwocks, whereas 'mathematics' deals with only some of the above. This means that the rules of logic, as we discuss them in this chapter, have an interest that extends far beyond mathematics and into other realms of human thought. Indeed, the subject of 'logic' didn't originate just from a desire to understand mathematical proofs, but from a wish to elucidate the general principles of correct reasoning in whatever area of life they were applied.

Perhaps a non-mathematical example will help make clear where we're heading. From the ideas of Chapter 3, we know that a statement like 'All swans are birds' can be expressed by a formula such as

$$\forall x \quad \text{Swan}(x) \Rightarrow \text{Bird}(x)$$

and the statement 'Anything that sees a swan sees a bird' by the formula

$$\forall x \quad (\exists y \quad \text{Swan}(y) \,\&\, \text{Sees}(x,y)) \;\Rightarrow\; (\exists y \quad \text{Bird}(y) \,\&\, \text{Sees}(x,y))$$

(We've omitted some brackets here, as we shall often tend to do from now on.) In the present chapter, our concern will be with the mechanics of the rules that enable us to deduce the second of these statements from the first.

I should admit at the outset that the slogan "The Rules of Thought" is perhaps a bit too grandiose for what we're doing here. We're really only concerned with what is known as *predicate logic*: that is, the logic that governs our six logical glyphs. We saw in Chapter 3 that these give all the logical structure we ever need for doing mathematics, and for quite a lot else besides. But it's far less clear that they suffice for everything that might go under the label of 'Thought' (even if we take the word to refer only to clear, precise, logical thought). We glanced briefly in Chapter 3 at some of the things going on in human language that seem to invite the use of other logical constructs, and these 'other logics' typically come with their own rules of deduction (which we won't attempt to describe in this book). Nonetheless, *predicate logic* is still an excellent starting point, even for the study of human language, since it gives us a stable and well-established basis on which to build. We also noted in Chapter 3 that even these 'other logics' typically have

an intended meaning that can be explained mathematically using just the constructs of predicate logic (possibly with some added non-logical machinery) — which raised the question of the question of whether our six logical operations do, after all, enjoy some fundamental place at the core of all human thinking. Perhaps future developments will shed further light on this.

For the rest of this (rather lengthy) chapter, we'll proceed as follows. After a quick look at some relevant history, we'll revisit the system of logical rules that featured in our story, and explain in more detail how they work. We'll then follow this up with a lighter interlude consisting of a puzzle about jabberwocks, before examining again the remarkable idea behind Gödel's *Completeness Theorem*. After that, things will get more philosophical. We'll explore in some depth an intriguing possibility that's opened up by our system of rules: namely, that of proving mathematical theorems without any reference at all to what they 'mean'. This idea is representative of an attitude to mathematics known as *formalism*, which has played an important shaping role in the development of the subject, and which has also given rise to some specifically 'formalist' philosophies of mathematics. We shall discuss all of this in some detail, introducing a number of key ideas which, in one guise or another, will remain with us for much of the rest of the book.

Logic through the ages

The attempt to isolate and codify the 'laws of thought' — to understand the rules that govern the process of correct logical thinking — has a long and venerable history. The subject of logic (as it concerns us here) was effectively originated single-handedly by the Greek philosopher Aristotle in his book, *Prior Analytics*, which described several patterns of valid logical deductions known as *syllogisms*, broadly similar in character to the Rules of Thought in our story. Here's a schematic representation of one of Aristotle's rules:

$$\frac{\text{All A are B} \qquad \text{No C are B}}{\text{No C are A}}$$

Here I'm following a modern habit of writing a rule of deduction using a long horizontal line: the statements above the line, known as *premises* (also spelt *premisses*), are the facts we suppose we know already, and the single statement below the bar (the *conclusion*) is something new that we can deduce from these. For example, if we already know 'All logicians are lazy' and 'No farmers are lazy,' then the above rule lets us deduce 'No farmers are logicians.' Aristotle's book collected and classified 14 rules of this kind, along with some further reasoning tactics called *conversion* and *contradiction*, and

showed how all these ingredients worked together to produce logical arguments.

What I find particularly striking is that Aristotle was interested not only in proving things *within* his logical system but also in proving things *about* it. For instance, he noticed that nothing new can ever be deduced from two negative statements, and also that one can in principle dispense with all but two of the syllogistic rules, since the other 12 can be 'proved' from these with the help of conversion and contradiction. This interest in looking at logic from the 'outside' as well as the 'inside' is very much in tune with the spirit of modern mathematical logic, and with the approach we're taking in this book.

Considered as a one-man effort at basically creating an entire subject from scratch, Aristotle's system was a great start, but it was nevertheless very limited in its scope. Specifically, it was only able deal with statements of four very simple forms:

All A are B Some A are B Not all A are B No A are B

So we can't use this system, as it stands, to get from 'All swans are birds' to 'Anything that sees a swan sees a bird', simply because the latter statement is of a more complex form than those that Aristotle's logic caters for.

Aristotle's rules were enthusiastically embraced by the medieval scholastics, who commentated on them at length and gave names to them all (the one I've shown above is called 'Camestres'). What's perhaps surprising, however, is that logic in the West advanced rather little beyond this stage until the 19th century: indeed, as late as 1800, the philosopher Immanuel Kant even claimed that nothing significant had been added to Aristotle's work on logic in 2000 years. It's worth noting that in the Islamic world, things did progress further: the remarkable Persian scholar Ibn Sīnā (nowadays remembered chiefly as a pioneer in medicine) extended Aristotle's system to cover some more complex forms involving 'or' and 'if-then' — not to mention building in a sophisticated form of 'time-sensitivity' into his logic. Regrettably, however, this work never became widely known within the European tradition.

The next major advance was made by George Boole who, in his book *An Investigation into the Laws of Thought* (1854), presented a system of logic that could deal with all statements built up using the so-called 'propositional' operations: in our notation, $\neg$, &, $\vee$ and $\Rightarrow$. In contrast to all earlier systems, Boole's algebra allows you to apply these operations repeatedly to build up statements as complex as you like. As we noted in Chapter 3, this system is powerful enough to form the logical basis of today's computer hardware; it also, incidentally, provides all the logic you'll ever need for puzzles such as Sudoku. However, Boole's treatment of $\forall$ and $\exists$ — the so-called *quantifiers* — was still quite limited and didn't really improve on Aristotle's.

It was Gottlob Frege in 1879 who first provided, in effect, a system of deduction rules for the whole of *predicate logic* — that is, for all six of our logical operations — a system powerful enough in principle for all logical reasoning in traditional mathematics. Frege's system was later tidied up and refined by Russell, Hilbert and many others, and many different ways of presenting essentially the same logic are now known. The one I'm using in this book is a (slightly idiosyncratic) variant of an idea known as *Natural Deduction*, introduced by Gerhard Gentzen in 1934; experts will notice a few points where I've departed from tradition in an attempt to keep the presentation as simple as possible.

Frege's work represented a huge milestone and in a sense opened up the whole field of modern mathematical logic. Even so, it seems highly unlikely that this is the last word on the basic question: *What is logic?* Not only does much of the 'logic' implicit in human language still await clarification, as we've noted earlier but even for classical predicate logic and its relatives, there may be scope for a deeper mathematical account of what a 'proof' really is: one that's less tied to seemingly artificial systems of symbols and rules. For the time being, though, it's through these clunky symbolic systems that the most accessible path to a grasp of logic leads, so let's now roll up our sleeves and take a closer look at one of them.

A system for classical predicate logic

When can a glyph statement be said to follow from other glyph statements by 'pure logic'? This is the basic question which the system of rules portrayed in the story is designed to answer.

The Rules of Thought aren't especially difficult, but they may appear complicated simply because there are rather a lot of them. This, in turn, stems from the fact that we have six different logical operations to deal with. Just as in the game of chess there are separate rules for the movements of kings, queens, knights and so on, so in the game of logic we have to give separate rules for 'and', 'implies', 'for all' and the others. On balance, I think the complete rules of predicate are probably of about the same complexity with the complete rules of chess, but that's still quite a bit to take in. I could, admittedly, have got away with a slightly smaller system, for instance by choosing to *define* some of the six logical operations in terms of others. However, I've resisted this temptation, since our goal is to gain some insight into the nature of these operations as they appear to our understanding, and declaring at the outset that some were defined from others would prejudge the issue. Overall, I've opted for the version which I think offers the best understanding on a first encounter.

Of course, for the purpose of grasping the main ideas, it's only the general *character* of these rules that matters, not the details; but on the other hand, there isn't really a better way of appreciating this general character than by looking at the actual nuts and bolts of some rules. There is also a certain satisfaction to be had from grasping a complete system that embodies so much of the richness and complexity of mathematics itself (in a sense we'll clarify later on). I've therefore decided to go through all of the rules here, including the ones we didn't look at in the story. This will take a bit of time, but we'll pick up speed as we go along, because once we've set up some basic notation for writing rules, it becomes easier to describe new ones. Even so, the details aren't too essential — so once you've had your fill of these rules, feel free to let the rest wash over you.

In the story, we saw some examples of the rules in action, but to describe cleanly how they're supposed to work *in general*, a bit of special notation will be helpful. In particular, I shall need to start by explaining how certain kinds of *variables* are going to be used.

We're familiar already with the idea that glyph formulae can contain variables (like x and y) which are intended to stand for numbers. But now we need to introduce variables of quite another sort: variables that stand for glyph formulae themselves! Just as we might say "Consider a number x," so we want to be able to say "Consider a glyph formula P." I shall use small letters (like x) for ordinary number variables, and capital letters (such as P, Q, R) for these new 'higher level' variables. It's important to notice that whereas the number variables are themselves part of the glyph language, the new higher level variables aren't: rather, they're part of the language that we ourselves are using to talk *about* the glyph language. Such an 'outer' level of language is often referred to as a *metalanguage*, and variables like P and Q that live at this outer level are called *metavariables*. This distinction between two levels of language may seem a bit confusing at first, but as you'll see, the use of metavariables will enable us to present our rules in a pleasingly compact form.

There's a second kind of metavariable that we'll need to get started. Remember from the story that a typical glyph theorem might look something like

$$4 \leq x\,,\ 5 < y \ \odot\ 9 < x + y$$

where we're allowed just a single glyph formula to the right of the medallion, but a *list* of formulae (just two in this case) to the left, playing the role of 'assumptions'. So it will be handy to introduce a metavariable **A** (in boldface) to stand for a list of glyph formulae, in the same way that P is used to stand for a single glyph formula. With this notation, we can now say that our *glyph theorems* — the things we're hoping to prove — will always have the form:

$$\mathbf{A} \odot P$$

For example, **A** might be the list $4 \leq x$, $5 < y$ and P might be $9 < x + y$, in which case we get the theorem displayed above. You can read such a theorem as saying 'assuming all the formulae in **A** hold, P also holds.'[1] Notice that the list **A** here might contain many formulae, or just one, or even none at all. In the last case, the theorem just says that P is true outright: no assumptions are involved.

Although every glyph theorem will have the form $\mathbf{A} \odot P$, not everything of this form will be a glyph theorem: for example, we certainly don't want there to be such a theorem as $\odot\ 0 = 1$. Our task, then, is to give a set of valid rules for creating glyph theorems, and the process of generating a theorem according to these rules will then be a 'proof' of that theorem.

Most of our rules will give us ways of creating a new theorem from previous theorems by means of a single step of 'deduction'. However, as Anna noticed, there'd better be some way of getting started — some way to produce theorems that doesn't depend on having other theorems already. Our first rule, corresponding to what the medallions that fell from the ceiling did, provides one way of doing this.

Assumption Rule:
If the formula P appears in the list **A**,
then $\mathbf{A} \odot P$ is a valid theorem.

In other words, if P is among the things we're choosing to assume, then we're certainly entitled to assert that P holds. So a theorem generated by this rule will just say 'If P and possibly a bunch of other things hold, then P holds.'

Actually, there's another way in which a proof might 'get started': we might import some glyph theorems from 'outside the temple' which aren't themselves deducible by pure logic, but rather contain some specific information about Whole numbers (or about Real numbers, or cockatoos, or whatever). In the next chapter, we'll look at the question of finding a suitable set of starting theorems, or *axioms*, for the theory of Whole numbers. But we'll leave this question to the side for now, since in this chapter we're focusing in what can be proved by pure logic alone.

As a brief digression, there's another rule similar to the Assumption rule that made a passing appearance in the story: the so-called *Weakening Rule,* which says that it's always permitted to add extra assumptions to an existing theorem. As Emily suspected, this isn't such an essential rule: indeed, it's quite easy to show that any theorem you can prove with the help of Weakening can also be proved without it, so it makes no fundamental difference whether we include this rule or not. For our purposes, it seems best

not to include it officially in our system, mainly because its use isn't recorded in our proof trees. But you could have such a rule if you wanted it.

All the remaining rules will give us ways in which theorems give birth to other theorems. In the story these were portrayed in terms of wooden branching structures, but here I'll write them using the horizontal line notation we saw earlier. The idea is that if we've already managed to create all the theorems that go above the line, the rule will generate the new theorem below the line. To see how this works, let's look at the general forms of the three rules for '&' (these were the first rules we met in the story).

& Intro:	**& Elim 1:**	**& Elim 2:**
$\mathbf{A} \odot P \quad \mathbf{A} \odot Q$	$\mathbf{A} \odot P \,\&\, Q$	$\mathbf{A} \odot P \,\&\, Q$
$\mathbf{A} \odot P \,\&\, Q$	$\mathbf{A} \odot P$	$\mathbf{A} \odot Q$

Hopefully the intuitive meaning of these rules should now be clear. For instance, the rule '&Intro' says that if the assumptions in **A** suffice to ensure that P holds, and also to ensure that Q holds, then these same assumptions suffice to ensure that P & Q holds. Notice that to use this rule in practice, we'll need to make some specific choice regarding what the metavariables **A**, P and Q stand for. For instance, when Anna and Emily first applied this rule, the two premises were the theorems $\odot\ 3 < 4$ and $\odot\ 0 \leq x$. This means, in effect, that they were taking **A** to be the *empty* list (there were no assumptions involved), P to be $3 < 4$, and Q to be $0 \leq x$. In this case, the rule delivers the new theorem $\odot\ 3 < 4\ \&\ 0 \leq x$. A more complicated instance of the rule was used by Anna in the proof tree shown in the story: we can obtain this by taking **A** to consist of the single assumption $3 < x\ \&\ y = 2$, P to be the formula $y = 2$, and Q to be $3 < x$. In this way, you can think of the general '&Intro' rule above as a kind of template from which an unlimited number of particular instances can be run off as we need them. The rules '&Elim1' and '&Elim2' can be understood in a similar way.

Let's turn now to the rules for $\Rightarrow$ ('implies'). Now that we've set up our general notation, the formal rules will hopefully speak for themselves without too much commentary:

$\Rightarrow$ Intro:	**$\Rightarrow$ Elim:**
$\mathbf{A}, P \odot Q$	$\mathbf{A} \odot P \Rightarrow Q \quad \mathbf{A} \odot P$
$\mathbf{A} \odot P \Rightarrow Q$	$\mathbf{A} \odot Q$

The '⇒Intro' rule is of particular interest because it makes significant use of assumptions. It says that if Q follows from P along with some other assumptions (as recorded in **A**), then we're allowed to conclude, just from those other assumptions, that P *implies* Q. Here we're writing **A** , P to mean some assumption list **A** with P tagged on to the right hand end. (If **A** happens to be the empty list, then **A** , P will refer to the list containing just P, even though there's no actual comma in this list.) The idea of this rule is that **A** , P can stand for *any* list ending with P, and the rule says that this rightmost P can jump over to the ⊙ symbol and combine with Q using '⇒'. Correspondingly, the '⇒Elim' rule tells us that relative to any list of assumptions **A**, we can combine P ⇒ Q with P to conclude Q.

The rules for & and ⇒, together with the Assumption Rule, are enough to let us prove some simple theorems liked the one proved Anna in Error: Reference source not found:

$$\odot\ (3 < x\ \&\ y = 2)\ \Rightarrow\ (y = 2\ \&\ 3 < x)$$

The proof tree, in effect, records the history of how this theorem is constructed: we use the Assumption Rule to get started with the theorems at the 'leaves' of the tree, then apply various rules in turn until the theorem at the root of the tree is reached. You may find it helpful at this point to look at this tree once more, and check that you can see how each step in the construction arises from the formal rules we've described.

Having got this far, it's now straightforward to give the rules for ∨ ('or') and for ¬ ('not'). No new notation is needed: the rules can simply be read in exactly the same way as those above. The rules for ∨ are:

∨ **Intro 1:**	∨ **Intro 2:**	∨ **Elim:**
$\mathbf{A} \odot P$	$\mathbf{A} \odot Q$	$\mathbf{A} \odot P \vee Q \quad \mathbf{A}, P \odot R \quad \mathbf{A}, Q \odot R$
$\mathbf{A} \odot P \vee Q$	$\mathbf{A} \odot P \vee Q$	$\mathbf{A} \odot R$

The Intro rules here are unsurprising: 'P or Q' can be deduced from P, and equally well from Q. The three-premise Elim rule, on the other hand, requires a bit of mental gymnastics: relative to some common background assumptions **A**, it says that if R follows from P, and R equally follows from Q, and we know that P or Q is true (though we may not know which), then we can conclude that R must be true either way. This is the way in which an occurrence of ∨ may get 'consumed' in order to prove something else.

As for ¬, the rules I shall give here unfortunately don't fit into our neat Intro-Elim scheme (this is a small casualty of my attempt to keep things

simple), but they're easy nonetheless. Once again, we already have all the notation we need, so here goes:

Contradiction:

$$\frac{\mathbf{A}, P \odot Q \qquad \mathbf{A}, P \odot \neg Q}{\mathbf{A} \odot \neg P}$$

Double Negation:

$$\frac{\mathbf{A} \odot \neg \neg P}{\mathbf{A} \odot P}$$

The Contradiction rule captures the principle that Euclid applied in our very first proof in Chapter 1: if some assumption P leads us to two contradictory conclusions (Q and not-Q), then we can conclude that P is false after all: in other words, not-P holds. Actually, the rule says that this principle can be applied even in the presence of other background assumptions represented by **A**. Of course, if **A** together with P leads to a contradiction, we don't in general know whether it's P or one of the assumptions in **A** that's wrong; but if we want to continue assuming **A** for the time being, we have no choice but to deny P.

The Double Negation rule says in effect that two negatives make a positive, so from not-not-P we may conclude P. This might seem harmless enough, but there is actually more to this innocent little rule than meets the eye, as we shall learn in Chapter 11.

That leaves just ∀ ('for all') and ∃ ('there exists'). These are perhaps the hardest part of the system to grasp, and they call for a little bit more notation. Rather than diving in with the general forms of these rules, we'll start with a special case of ∀ Introduction, taking '$x \leq x$' as an example to work with.

$$\frac{\mathbf{A} \odot x \leq x \qquad\qquad x}{\mathbf{A} \odot \forall x \;\; x \leq x} \quad (x \text{ not free in } \mathbf{A})$$

Let's start with the mysterious side-condition 'x not free in **A**.' Remember that x is an ordinary variable which we may think of as standing for a Whole number. The intention behind the side-condition is to capture the idea that the assumptions in **A** don't say anything at all about the value of x. Broadly, the idea is that if it's possible to know '$x \leq x$' without knowing or assuming anything about x, then we're in a position to conclude that this statement holds for *all* values of x: that is, $\forall x \;\; x \leq x$.

The exact meaning of 'x not free in **A**' is rather technical. Certainly this condition will be satisfied if the variable x doesn't appear within **A** *at all* — in this case, it's very clear that **A** can say nothing about the value of x. In general there's a little bit more to it than this, but I'll relegate that to an endnote.[2] If

you prefer, you can just read 'x not free in **A**' as saying 'Terms and Conditions Apply'.

You can see that, as you'd expect of a '∀ Intro' rule, this rule introduces a ∀ into the proceedings where there wasn't one before. Of course, every ∀ needs an associated variable, in this case x, and the way I've set up the rule allows you to supply your choice of variable as an extra 'input' to the rule. This is needed because in general there might be several variables around: for instance, if in place of $x \leq x$ we had the formula $x \leq x \,\&\, y \leq y$, we'd want to be able to choose whether we introduced '∀x' or '∀y'.

We've been looking so far at a special case of ∀ Intro, but let's see now how it can be generalized. Firstly, in place of $x \leq x$ we want to allow an arbitrary formula P, which will typically one containing the variable in question (x in the above case). So we should replace both the occurrences of '$x \leq x$' in the above rule by 'P'. But secondly, as already indicated, we want to allow other variables like y or z to play the role of x here. To capture this, we use a notational trick: just as we've introduced a metavariable P to stand for a glyph formula such as $x \leq x$, so we will now introduce a new metavariable *v* which may stand for any ordinary variable such as x, y or z. (The use of italics here is significant: a plain v would be just another ordinary variable.) Armed with this convention, we can frame the general rule as follows.

∀ Intro:

$$\frac{\mathbf{A} \odot \mathrm{P} \qquad v}{\mathbf{A} \odot \forall v \ \mathrm{P}} \quad (v \text{ not free in } \mathbf{A})$$

Now for the ∀ Elimination rule. Once again, it's helpful to start by looking at a specific example. Suppose we'd already proved $\forall x \ \ x \leq x$ (with the help of some assumptions). What would we expect to be able to deduce from this? Since we've shown that $x \leq x$ for *all* x, we'd certainly expect to be able to conclude that this holds for any *particular* choice of x we're interested in: for instance, we could conclude that $3 \leq 3$, or that $4 + 5 \leq 4 + 5$, or that $y \leq y$ (whatever the value of y), or that $(z + 2) \times 7 \leq (z + 2) \times 7$. We can capture this idea as follows. We'll introduce another metavariable, T, which can stand for any *term* in the glyph language: something like 3, 4 + 5, y or (z + 2) × 7, which we would normally understand as representing a number. We can then express all the above instances of ∀ Elimination as follows:

$$\frac{\mathbf{A} \odot \forall x \ \ x \leq x \qquad \mathrm{T}}{\mathbf{A} \odot \ \mathrm{T} \leq \mathrm{T}}$$

Here, the term T is supplied as an additional input to the rule in order to 'tell' it what we want the variable x to be specialized to.

We'd now like to generalize this rule so as to allow any formula P to take the place of $x \leq x$. But this raises a question: if we replace '$x \leq x$' by 'P' in the rule above, what should we replace '$T \leq T$' by? What is the precise relationship between $x \leq x$ and $T \leq T$? Clearly, the answer is that $T \leq T$ is what we get from $x \leq x$ by *replacing* both occurrences of x by T. This suggests that a bit more meta-level notation might be useful: if P is any formula, we can write $P[x \to T]$ for the formula obtained from P by replacing occurrences of x by T. Actually, there's another bit of technical smallprint here, which again I shall tastefully conceal in an endnote[3] — but broadly, you can think of $P[x \to T]$ as 'P with x replaced by T everywhere'.

There's one other generalization we want to make: we want to allow any other variable like y or z to play the role of x. Once again, we can use our metavariable *v* for this purpose, leading to the following general rule:

∀ Elim:

$$\frac{\mathbf{A} \odot \forall v \; P \qquad T}{\mathbf{A} \odot P[v \to T]}$$

Both our rules for ∀ took quite a bit of work. But having developed the necessary notational machinery, it's now easy to give the rules for ∃ as well:

∃ Intro:

$$\frac{\mathbf{A} \odot P[v \to T] \qquad v \qquad P}{\mathbf{A} \odot \exists v \; P}$$

∃ Elim:

$$\frac{\mathbf{A} \odot \exists v \; P \qquad \mathbf{A}, P \odot Q}{\mathbf{A} \odot Q} \; (v \text{ not free in } \mathbf{A} \text{ or } Q)$$

The ∃ Intro rule says that if we've established even just one instance of a formula like $x \leq x$ — say we know that $z + 5 \leq z + 5$ — then we can conclude that $\exists x \; x \leq x$. The small technical point here is that in order for the rule to 'know' which conclusion we're wanting, we need to supply both a choice of variable *v* (in this case x) and a choice of formula P (in this case $x \leq x$). If we used the same premise asserting that $z + 5 \leq z + 5$, but instead took *v* to be y and P to be $y \leq z + 5$, we could legitimately draw a completely different conclusion: $\exists y \; y \leq z + 5$.

Finally, the ∃ Elim rule involves a mental twist rather similar to ∨ Elim: roughly, if a property P, which may say something about x, suffices to ensure

some property Q that says nothing about x, then the mere existence of *any* x that satisfies P will be enough to ensure that Q holds. (If there are other assumptions involved, these had better not say anything about x either, or the argument will break.) In line with the other Elim rules, this gives us a way of using knowledge of an ∃ statement to prove something else.

A universal, reductive, mechanizable system

That concludes our presentation of the logic: congratulations if you got all the way through it! Now that all the pieces are in place, we're in a position to step back and note some important aspects of what we've just done.

First, all the rules we've given can be considered as rules of 'pure logic': they deal only with the six logical symbols, and there's no mention anywhere of non-logical symbols like 5, + and <. Indeed, although we've implicitly been supposing that all the formulae featuring in our rules are formulae of Arithmetic, this doesn't have to be the case: exactly the same rules could be understood as applying to any other language with the same logical symbols but a different non-logical vocabulary: say with symbols appropriate to geometry, set theory or jabberwock theory. (We can refer to any such language as a *dialect* of glyph language.) It's in this sense that our rules can indeed be viewed as universal rules of logic.

Secondly, since it's clear how to combine our rules to form proof trees, our system gives us a precise working definition for what a 'proof' is. So in this sense, we've answered the question we raised at the start of the chapter, at least as far as proofs in pure logic are concerned. What's more, this definition of proof has a pleasingly 'reductive' character: the task of checking whether some big proof is correct is broken down into a large number of much smaller tasks that can be carried out independently, namely the tasks of checking that the individual rule applications are correct. Beyond this, there's no further condition on the proof *as a whole* that needs to be checked — for instance, on the way two distant parts of the tree relate to one another. (Contrast this with chess: you can't *quite* tell whether a chess game is legal just by looking at each move separately, because there are a few 'non-local' rules: for instance, either player may claim a draw if the same game state occurs three times.) Another good property is that the rules for the six logical operations can themselves be understood quite independently of one another, since no rule involves more than one of these symbols. (In other words, there's nothing akin to the *castling* move in chess, which involves pieces of two different kinds.)

Thirdly, each of the individual rules of our system is simple enough to be mechanizable. That is, for each of our rules, it would be quite easy to build a computer program (or tube machine if you prefer) that took as 'inputs' whatever we choose to put on top of the rule — the premises and any other

information such as a choice of variable — and delivers as 'output' the new theorem that the rule produces. This means that we can use a computer to help us construct proofs in such a way that the correctness of each step is mechanically verified: the idea is that the human user still guides the computer through the proof by telling it which steps to apply when, but the computer actually performs the steps, checking their legitimacy and perhaps filling in some boring intermediate steps too.

This idea has indeed been made the basis of *theorem-proving assistants* such as HOL, Isabelle, Coq and Agda (of which the first two use logical systems quite similar to the one we've described). For some people, the value of these systems lies in the possibility fully machine-checked proofs of mathematical theorems: this has proved particularly valuable in some cases where the proofs are so complex that it's hard otherwise to be confident that they don't contain a mistake.[4] For other people, the primary interest is in using these systems to give machine-checked correctness proofs for hardware or software systems. Either way, it is thanks to the fundamental logical work of Frege and others that these developments are possible.

At this point, it's possible that you may want to try working out some formal proofs for yourself. If so, I've provided some suggested exercises in a supplement, *Formal proofs in pure logic*, on the book's website. One word of warning, though: the 'proof tree' format that I've presented, although it suits the purposes of this book well, isn't the most convenient when it comes to actually writing out proofs. This is partly because of the difficulty of finding a piece of paper wide enough for any decent-sized proof tree, and partly because of the endless copying out of assumptions that it entails. For the purpose of working out particular proofs, I'd recommend a slightly different format using exactly the same logical rules, but a more economical way of laying things out. The style I have in mind is called *Fitch notation*, and this is explained (with examples) in the supplement mentioned above.

One thing that's admittedly rather hard to convey by means of small examples is any sense that our system is capable of proving anything really *interesting*: most 'small' proofs that are easy to construct result only in rather insipid logical truisms. In fact, this is only a question of scale: any given part of a complex mathematical proof will indeed look trivial if you zoom in to the micro-level of individual rule applications, just as the most stunning digital image will turn out to consist of nothing but pixels if you zoom in far enough. But by combining enough of these trivial steps, it turns out that one really can conduct non-trivial arguments. Or to put it another way: for someone accustomed to ordinary 'human' mathematical reasoning, being forced to do a proof within our system is rather like being shrunk down to the size of an ant: tiny bumps on the ground that you'd normally have stepped over without noticing suddenly become huge mountains to be arduously traversed. I first encountered this experience for myself when working with the HOL theorem-

prover about 30 years ago, but the experience is still familiar to researchers even with today's tools: fully formal proofs are often extremely laborious to construct.

We also made the point earlier that to do genuine mathematical proofs, we typically need a bit more than a system of pure logic: we also need some non-logical axioms giving the basic mathematical facts we're starting from. However, for a typical mathematical proof, the main difficulty lies not in choosing which axioms to allow, but in proving that the desired theorem logically follows from these axioms. In this sense, most of the richness and complexity of existing mathematics is indeed present in the potentiality of the system we've described. It's also worth remarking that, axioms apart, the general question of whether a given glyph statement is provable by pure logic turns out to be *undecidable* in the Turing sense — a clear indication of the complexity inherent in our system.

As a fairly simple example to illustrate that our system really can express 'non-obvious' kinds of reasoning, here is a little logic puzzle about jabberwocks.

The theory of jabberwocks

There are just two things you need to know about jabberwocks. First, there are at least six of them. Second, certain jabberwocks may be said to *whiffle* certain other jabberwocks: that is, if a and b are jabberwocks, then either a whiffles b or it doesn't. Your challenge is now to show that there are three distinct jabberwocks a, b, c for which one or other of the following statements is true:

1. a whiffles b, b whiffles c, and a whiffles c.
2. a doesn't whiffle b, b doesn't whiffle c, and a doesn't whiffle c.

You might enjoy trying to figure out by everyday logic-puzzle reasoning why this is so. But my main purpose here is to show how we can express the problem in our logical system. For this purpose, we'll use a dialect of glyph language with just two kinds of basic formulae: a = b and a 🐍 b. The intended interpretation is that variables stands for jabberwocks, a = b means 'a is the same as b', and a 🐍 b means 'a whiffles b'.

We'll need a couple of formal assumptions to capture the idea that there are at least six jabberwocks. Consider the following two formulae:

$$\exists a \quad a = a$$

$$\forall a \ \forall b \ \forall c \ \forall d \ \forall e \ \exists f \ (\neg f = a \ \& \ \neg f = b \ \& \ \neg f = c \ \& \ \neg f = d \ \& \ \neg f = e)$$

The first formula is just a way of ensuring that there's at least one jabberwock. The second formula says that if a, b, c, d and e are any jabberwocks (not necessarily different), then there's some jabberwock f that's not the same as any of these. We'll write **A** for the list consisting of just these two assumptions. Notice that there's no mention of whiffling at this stage. Notice too that we haven't assumed (and actually don't need to assume) that a = b implies b = a.

A first exercise is to try to see, at least in outline, how one would use the assumptions of **A** to prove that there are at least two jabberwocks: more precisely, how to prove the following theorem in our system:

$$\mathbf{A} \odot \exists a \;\; \exists b \;\; \neg b = a$$

Having done this, we can apply the same idea again to show that there are at least three jabberwocks in the following sense:

$$\mathbf{A} \odot \exists a \;\; \exists b \;\; \exists c \;\; (\neg b = a \;\&\; \neg c = a \;\&\; \neg c = b)$$

And so on, until we reach a rather long and cumbersome formula saying there are at least six jabberwocks.

But all of that is really a side issue. The main challenge is to see how to prove the following theorem:

$$\mathbf{A} \odot \exists a \;\; \exists b \;\; \exists c \;\; \neg b = a \;\&\; \neg c = a \;\&\; \neg c = b \;\&\; ((a \curlywedge b \;\&\; b \curlywedge c \;\&\; a \curlywedge c) \vee (\neg a \curlywedge b \;\&\; \neg b \curlywedge c \;\&\; \neg a \curlywedge c))$$

which is a way of expressing the goal I stated at the start of this section. This is quite a good example of a problem in 'pure logic', since it invites us to prove something non-obvious about whiffling from assumptions that don't mention whiffling at all.

I certainly don't recommend that you try to write out a complete formal proof of the above theorem — it would be very large! But it can be done, and you may enjoy thinking it through to the point where you're convinced of that. A solution to the puzzle, and an outline of how a formal proof can be constructed, are included in the supplement on *Formal proofs in pure logic*.

Gödel's completeness theorem

We've just been looking at what we can prove *within* our system. But now let's change gear and see what one can prove *about* it.

Once we've set up our system of 'Rules of Thought', it's very natural to ask is whether we might have left anything out. Could there be other purely

logical principles that we've simply forgotten to add? Or even other universally valid principles of reasoning which aren't accessible to our human intellects, but might be accessible to an alien super-intelligence?

The *completeness theorem* shows that as far as (any dialect of) our glyph language is concerned, the answer is no: anything that could reasonably be considered a 'truth of pure logic' can be proved using just the rules we've given. This remarkable result was first obtained by the young Austrian logician Kurt Gödel in his doctoral dissertation of 1929 — just a year before his more famous *incompleteness theorems*, which we'll come to in the next chapter. Although we've already covered the completeness theorem in the story, the idea is quite subtle, and worth going over one more time.

How might we *prove* that every glyph statement that's 'true by virtue of logic alone' can be proved using our rules? The problem, on the face of it, is that we don't have an independent definition for what 'true by virtue of logic alone' means. Fortunately, we don't need one. All we need to agree on is that if a statement is 'true by logic alone' in *any* reasonable sense, its truth won't be dependent on the specific meanings we assign to the non-logical parts of the language. If the statement were true for Whole numbers but false for meringues, say, then it would say something we couldn't know without knowing some specific property of Whole numbers that wasn't shared by meringues — in other words, it wouldn't be something that was 'true by logic alone'. Indeed, even *one* interpretation of the non-logical symbols that made the statement false would be enough to show that it didn't have this charmed status. So if we can prove:

Any statement that's true however the language is interpreted
can be proved using our Rules of Thought

then that should be enough to satisfy us that

Any statement that's true by logic alone (in any reasonable sense)
can be proved using our Rules of Thought.

How can we make precise this idea of a 'however the language is interpreted'? This leads us to the concept portrayed by the balloons in our story: the concept of a *model* for our glyph language. A model is basically a kind of 'field of reference' which the language can be construed as talking about. To give a model, we need to specify two ingredients. First, some collection of *Things* (e.g. Whole numbers, meringues, jabberwocks) which we can consider the language to be talking about. The variables in our language will then be taken to stand for these Things. Second, some way of interpreting the various symbols (e.g. 0, 1, <, +, if our dialect contains these) in terms of these Things. How we do this will depend on the symbol in question. For

symbols like '0' and '1', it's enough to say which Thing we want the symbol to refer to. For a symbol like '<', we have to say which *pairs* of Things we want to consider as being in the '<' relationship. And for a symbol like '+', we need to give some description of how '+' can be applied to a pair of Things to produce a third Thing.

Once we have a model, we can proceed to say what it means for a glyph statement to be *true* for that model. This is very similar to what we did in Chapter 3, using Tarski's method for defining what is meant by 'true in the Whole numbers' or 'true in the Real numbers'. In fact, the Whole numbers and the Real numbers will be just two particular examples of models for our standard glyph dialect, and exactly the same Tarskian approach can be used to define what we mean by 'true' for any model we like.

The idea, then, is that any model offers a possible way to interpret the language. So being *true however the language is interpreted* should certainly imply being *true in every model*, and the latter is now a precisely formulated mathematical condition. This means that we will establish the fact we're interested in if we can prove the following mathematically precise statement:

Any glyph statement that's true in every model for the glyph language can be proved using our Rules of Thought.

And this is precisely what the Gödel completeness theorem tells us.

Actually, Gödel proved a bit more than this.[5] In addition to glyph statements that were 'true outright', he also considered the case of statements that followed from a set of *axioms* — perhaps even infinitely many — by logic alone. Although it's a bit more technical, it's worth glancing at this more general form, as it will prove quite handy in later chapters.

By an axiom, we shall mean any glyph theorem that we choose to allow 'for free' as a starting theorem at some leaf in a proof tree. For convenience, we'll insist that all our axioms are of the form ⊙ Q (the list of assumptions on the left being empty), where Q is a 'self-contained' formula with no free-floating variables. We'll use **X** to stand for any (finite or infinite) set of glyph axioms which are each of this kind. We can now state a more general form of the completeness theorem:

Let **X** *be any set of glyph axioms, let* P *be a single glyph statement.*
Suppose that in every model in which all the axioms in **X** *are true,*
P *is also true.*
Then ⊙ P *can be deduced from axioms in* **X** *using our Rules of Thought.*

In other words, if P 'follows from' the axioms of **X** in the sense that whenever these axioms are true, so is P, then P also 'follows from' **X** in the sense that it can be proved from these axioms. Of course, if the set **X** happens

to be empty, then the latter just means that P is provable by pure logic alone, so in this case we retrieve the simpler version of the completeness theorem as stated above.

The complete set!

The significance of Gödel's completeness theorem is perhaps a bit under-appreciated, as it tends to get overshadowed by his more shocking and sensational incompleteness theorems. But let's just pause to take stock of what the completeness theorem tells us.

We started with the old question, going back at least to Leibniz and perhaps to Aristotle: 'What exactly are the rules of correct logical thinking?' We've seen that Frege in 1879 made a key breakthrough resulting in both a precise formal language (as in Chapter 3) and a system of deduction rules (as in this chapter) for logical statements. Whilst Frege's language may directly capture only one part of the totality of humanly thinkable thoughts, it's a part that's large enough (with suitable vocabulary) to cover the whole of mathematics, and quite a lot else besides. Frege's system was improved by Russell and Hilbert; then in 1930, Gödel proved that the system is *complete*: for the class of statements that it covers, it's capable of all the logical deductions one could possibly hope for.

As far as it goes, this all adds up to a complete and satisfying answer to our original question: a comprehensive analysis of what we recognize as 'logical reasoning' (always with the caveat that we're limiting attention to what can be said in dialects of our glyph language). And given that such reasoning serves as the mortar for the entire edifice of mathematics, the discovery of its underlying laws, no less than the Church-Turing definition of mechanical computability, should surely be hailed as an intellectual achievement of the first magnitude. Leibniz himself eloquently expressed his sense of the high dignity of his endeavour:

> For if praise is given to those who have determined the number of regular solids — which is of no use, except insofar as it is pleasant to contemplate — and if it is thought to be an exercise worthy of a mathematical genius to have brought to light the more elegant properties of a conchoid or cissoid, or some other figure which rarely has any use, how much better will it be to bring under mathematical laws human reasoning, which is the most excellent and useful thing we have.[6]

Two centuries later, George Boole wrote in a similar vein:

> To unfold the secret laws and relations of those high faculties of thought by which all beyond the merely perceptive knowledge of the world and of ourselves is attained or matured, is an object which does not stand in need of commendation to a rational mind.[7]

In 1927, the discovery of the logical systems we're discussing was celebrated by David Hilbert with an enthusiasm echoing that of Leibniz and Boole:

> The formula game … has, besides its mathematical value, an important general philosophical significance. For this formula game is carried out according to certain definite rules, in which the *technique of our thinking* is expressed. These rules form a closed system that can be discovered and definitively stated. The fundamental idea of my proof theory is to describe … the rules according to which our thinking actually proceeds … If any totality of observations and phenomena deserves to be made the object of a serious and thorough observation, it is this one.[8]

Finally, Gödel crowned all this with a mathematical proof confirming that a complete set of such rules had indeed been found: there were no fundamentally new logical principles awaiting discovery.

So some major aspect of what it means to think rationally can be comprehensively captured by a set of precise symbolic rules. It's no exaggeration to say that this discovery is to logic what the discovery of the structure of the atom is to physics: a fundamental scientific analysis of the basic ingredients of the subject matter.

But what *is* logic?

So we now know quite precisely, in a certain sense, what 'logic' consists of. But does that mean we know what logic really *is*? We have given, so to speak, a detailed anatomy of logic, but have we really captured the life and soul of the beast?

We've seen, for instance, how logical reasoning can be analysed in terms of a set of fairly simple rules. Since these rules are individually mechanizable in character, there's no difficulty in principle in imagining how such rules (or something equivalent) might be built into our brain processes. So on one level, our analysis of logic shows that there's no particular mystery about our minds' ability to carry out logical reasoning. But as we saw in Chapter 1, that doesn't seem to be the whole story. For not only are we able to *apply* these rules as we can the rules of chess, but our minds are in some way able to *recognize* their absolute, unassailable validity — that quality of 'mustness' inherent in them. How does this work? How can our minds tell what's an

absolutely valid law of logic and what isn't, and why are we so sure that what our minds tell us is right? Why do we consider it *absolutely certain* that if cockatoos eat peanuts then either cockatoos eat peanuts or quetzals eat meringues — even if we've never seen either of these birds, and even if some eminent ornithologist should try to tell us otherwise?

What *kind* of thing is logic anyway? Where does it live? When we look at the logical rules I presented earlier, are we looking at a representation of some deep and essential property of ultimate reality — an immutable aspect of the way things objectively are? Or are we looking at a reflection of some feature of our own minds — at the ways in which we cannot but think about the world? Or merely at a system of cultural conventions governing the way we use certain words or symbols? Isn't there *some* sense in which it's not 'up to us' to decide what's a valid law of logic and what isn't?

These are deeper questions about the nature which our symbolic analysis of logical rules doesn't even touch on. Broadly, they have to do with the nature and status of what philosophers call *a priori* knowledge: that is, knowledge that our minds seem able to come up with 'all by themselves', independently of observing how things are in the external world. Exactly how such knowledge is possible is still a matter of intense philosophical debate.[9]

On the one hand, the whole of mathematics has always be seen as strong candidate for *a priori* knowledge — which is one reason why philosophers find mathematics so intriguing, as we saw in Chapter 1. I hope the discussion there persuaded you that the question of how we perceive mathematical truth is by no means an easy one to answer. But it's worth noting that even the humbler realm of purely *logical* truth has much the same status, and the same basic questions can already be asked in this setting.

We shall return to these deeper questions towards end of the book. But for the remainder of the chapter, I'd like to turn to another idea we touched on in our story — an idea which might, at first, seem to offer a neat way of sidestepping all these hard metaphysical questions. This is the idea of *formalism*.

The formalist attitude

Like many 'ism' words, formalism is an umbrella term covering quite a wide range of ideas. What I shall here take to be the essence of formalism is the general attitude that *the activity of doing mathematics* (or some of it) *can be in some way divorced from considerations of what it might mean*. We met this idea in the story: once we've formulated the laws of reasoning in a symbolic, mechanical fashion, it no longer matters, for the purpose of building proof trees, whether these statements are talking about numbers, jabberwocks

or indeed anything at all. We can just choose whatever formulae we like as initial axioms, then just blindly apply the rules in a purely symbolic way.

To my mind, formalism is best considered not so much as a philosophy of mathematics as a general *attitude* — one which can, to some extent, coexist with many other attitudes and philosophies. This attitude exists in a variety of shades and strengths, but it's generally manifested by such things as: (a) preferring to do mathematics within frameworks where the allowable modes of reasoning are rigorously spelt out by purely symbolic rules; (b) declining to engage with questions about what, if anything, mathematical statements might mean; and (c) being happy to work within mathematical frameworks to which no clear notion of 'meaning' has been attached. However, a more hard-line formalist might go a step further than this, and maintain that mathematical statements (or at least some of them) are actually just combinations of symbols that *don't have a meaning at all*. At this point, of course, formalism does indeed become a philosophy of mathematics.

Now, if you've spent a fair bit of your life actually doing mathematics, it's possible you may find this last idea repugnant, shocking, or downright silly. *Surely* mathematics is more than just manipulation of symbols? Surely the whole point of doing mathematics is to discover 'truth' of some sort? Isn't that the very essence of the mathematical creed? And anyway, why would anyone *want* to take the contrary view? Why leave the green pastures of truth and meaning for the arid wilderness of mere symbol-shunting?

In order to understand what motivates the formalist view, we shall need to look at the history of mathematics to see how and why this attitude arose, and what role it has played in various periods. This will help us to appreciate both the benefits that formalism has brought and the questions it leaves us with.

Formalism: an informal history

In the beginning was Euclid. As we saw in Chapter 1, Euclid's style emblematized the ideal of mathematical rigour: starting from a small set of clearly stated axioms which were considered to need no justification, he sought to develop his whole theory of geometry by means of logic alone. And although (as we saw) Euclid's development of geometry didn't entirely live up to this ideal, the ideal was certainly there.

For most of the next 2,000 years, however, mathematics wasn't usually practised in quite such a rigorous way. By and large, mathematicians showed little concern for 'laying the foundations' of what they were doing, and in the course of a proof it was generally deemed acceptable to help yourself to any 'immediately obvious' assumptions you needed as you went along. But the very notion of something being 'obvious' presupposes that it has a meaning

you understand. So in this sense, the acceptability of a mathematical argument was grounded in our *understanding* of what the argument was about.

Later on, though, new developments arose which led many mathematicians to adopt a more pragmatic attitude, and to admit certain kinds of reasoning that appeared to give the right answers, even without a clear idea of what it 'meant'. Two outstanding examples of such discoveries were the theory of *Complex numbers*, and that of the *differential calculus*.

Complex numbers seemed mysterious in the extreme when they were first mooted around 1550, and the idea still continues to baffle legions of school students, much as it baffled Emily when she met them in Chapter 2. The Real numbers, at least, are easy to 'understand' intuitively: we can grasp the idea of numbers that measure some continuously varying quantity such length or time. Then one day we're told: "Suppose there were a number (let's call it i) whose square is minus one." "But there *isn't* any such number," we protest. "Well, let's just pretend there is, and see where that leads us". "Okay, if you insist," we say, and we start pushing the symbols around according to the algebraic rules we've been taught. "But surely," we might murmur, "that means we're not talking about what's *true* any more? Aren't we just talking about a fairy story we've invented?" Indeed, that's just what many mathematicians *did* say, on and off, for over two hundred years.[10]

Nevertheless, no one could deny that these 'imaginary' numbers were extremely useful for solving an enormous range of problems. (A classic example is their use in solving *cubic equations*, such as $x^3 + 2x^2 + 3x + 4 = 0$. The method makes use of the algebra of Complex numbers, even in cases where the solutions themselves turn out to be ordinary Real numbers.) For this reason, most mathematicians were happy to embrace the use of Complex numbers, overlooking any nagging questions about whether such numbers really 'existed'.

Nowadays, incidentally, Complex numbers aren't seen by mathematicians as shrouded in any such mystery. If you want to say what a Complex number 'is', it's actually quite simple: you can say it's nothing but a *pair* of Real numbers. For example, the expression $2+3i$ can be seen as just a fancy name for the pair (2,3). We can then *define* particular operations on such pairs which we choose to call 'addition', 'multiplication' and so on. This is a simple idea, but it took a surprisingly long time for its power to be fully appreciated.[11] Before then, mathematicians in effect had to content themselves with a *formalistic* attitude towards the Complex numbers: 'We don't think the symbol i 'refers' to anything, but we know how to do the algebra.'

This pragmatic trend was even more strikingly manifested when Newton and Leibniz introduced the *differential calculus* around 1670. The calculus was a wonderful new technique for calculating the slopes of curves at specified points — a technique which worked brilliantly in practice, leading to nothing less than a revolution in the physical sciences. But the embarrassment

was that no one really understood *why* it was working so well, because — as Newton and Leibniz themselves were fully aware — the reasoning on which it was based was logically quite incoherent. Put simply, the reasoning depended on assuming the existence of numbers called *infinitesimals*: numbers that were in some way 'infinitely small' without actually being zero. Of course, on our usual understanding of Real numbers, there are no such animals at all. Just like the imaginary number i, infinitesimals were symbols that one could juggle with and get very useful results, even though they didn't seem to refer to anything. As Leibniz put it:

> To speak philosophically, I would advocate neither infinitely small nor infinitely large magnitudes … To put it compendiously, I hold these to be mental fictions, suited for use in calculations, like the imaginary roots in algebra. And yet I have demonstrated that these expressions are a great aid in shortening thought and also in discovery.[12]

This lack of any meaningful foundation for the differential calculus became the butt of a shrewd if rather waspish attack from one Bishop Berkeley, who argued that anyone who could swallow the theory of infinitesimals and all the contradictions it entailed 'need not, methinks, be squeamish about any point in Divinity.'

Because these new methods were so obviously powerful, mathematicians were eager to press ahead and see what they could do, much as an applied mathematician might routinely do today.[13] And as long as the calculus seemed to be leading to the right answers, one could indeed turn a blind eye to the defects in its foundations. However, as time went on, and mathematics grew more and more ambitious, the cost of this pragmatic attitude started to manifest itself. Mathematicians began to find that the symbol-juggling rules they were applying, whilst sometimes leading to sensible and useful results, at other times gave rise to sheer nonsense.[14] So how was one to know *which* applications of the rules could be trusted? It became a matter of urgency to sort out the mess, and to find ways of distinguishing legitimate uses of the rules from illegitimate ones. At this point (around 1820), the tide turned, and the trend thereafter became one of increasing rigour and precision — a trend that continued into the twentieth century and set the tone for pure mathematics as we know it today.

You might have expected that the effect of this new concern for logical rigour would have been a move *away from* the formalist attitude and a return to a more meaning-oriented approach to mathematics. What actually happened was perhaps surprising.

It was the German mathematician Karl Weierstrass who, from 1857 onwards, was chiefly responsible for putting the calculus on a logically coherent foundation. Weierstrass's approach banished all reference to

infinitesimals, replacing them with completely kosher statements about what would be true for sufficiently small 'ordinary' numbers. (This is the so-called 'epsilon-delta' approach now standardly taught in university maths courses.) This straightened out the confusion, clarified what was legitimate, and generally left the whole subject in a far better shape.

However, the newfound rigour of Weierstrass's approach led him to ask questions no one had asked before. For example, his logically meticulous treatment had forced him to give a precise definition of what exactly a *continuous curve* is — an idea which had been accepted as intuitively obvious till then. (The word 'curve' is used here in a broad sense, which covers straight lines and also things with sharp corners. The operative word here is 'continuous', which means that a curve mustn't have *jumps*. The basic intuition is that a continuous curve is something that can be drawn without the pencil ever leaving the page.) In the course of his work, Weierstrass came across examples of 'continuous curves' that were far from intuitive, and that flew in the face of what might previously have been thought 'obvious'. For example, picture a curve in your mind. You will probably be picturing something which, at each point, has a well-defined 'slope' or 'direction' — possibly with the exception of a smallish number of sharp 'corner points'. However, Weierstrass found examples of continuous curves that didn't have a well-defined slope at any point at all (some of these curves would be known nowadays as *fractals*). An even weirder example was due to Giuseppe Peano, who showed how to define a single continuous curve that 'filled up' an entire square, in the sense that it visited every single point within it.

These bizarre discoveries led to a severe loss of confidence among mathematicians regarding their ability to tell what was 'obvious'. Whereas in earlier times, mathematicians might have freely helped themselves to intuitively obvious facts as and when they needed them, now the very idea of something being 'obvious' was treated with the gravest suspicion, and there was a general retreat from the older idea that it was our *intuitive understanding* of a proof that provided the grounds for accepting it. The spirit of the age was summarized by the mathematician Hans Hahn:

> Because intuition turned out to be deceptive in so many instances and because propositions that had been accepted as true by intuition were repeatedly proved false by logic, mathematicians became more and more sceptical of the value of intuition … Thus a demand arose for the expulsion of intuition from mathematical reasoning and for the complete formalization of mathematics … every new concept was to be introduced through a purely logical definition; every mathematical proof was to be carried through by strictly logical means.[15]

In this way, the role of meaning in mathematics became gradually marginalized: even if you did have an 'understanding' of what you were talking about, you're weren't allowed to appeal to it in the course of a proof. You were merely performing purely logical deductions from explicitly declared axioms — and consequently it became a matter of indifference (logically speaking) what, if anything, you were talking 'about'. In geometry, for instance, you might use words like 'point' and 'line', but nothing essential depended on whether you knew what a point or a line *was*. David Hilbert, in his treatise on geometry in 1899, expressed the idea in a characteristically Germanic fashion when he said that in place of points, lines and planes, it must always be possible to speak of tables, chairs and beer mugs.

Eventually, with the work of Frege and others, even the rules of logic itself were codified in symbolic systems like the one we've been looking at, so that it truly became possible to regard mathematical proof as reducible, in principle, to pure symbol-shunting. In this way, Euclid's original ideal of a flawless logical edifice founded on basic axioms had returned with a vengeance, and was now being prosecuted with an austerity and obsessional rigour that the Greek master could scarcely have dreamed of.

But there's an obvious question here. If we're no longer allowed to appeal to what's 'obvious', how do we even know that our basic *axioms* are true? Isn't the whole point of axioms that they're sufficiently obvious not to need proof, and therefore give us a place to start from? That had been how they were seen in the past. But now that the very notion of being 'obvious' was subject to suspicion, people started questioning everything in sight, and wondering what (if anything) in could be considered as truly certain.

And from here, as you can imagine, it was only a small step to a point of view that said simply: We *don't* know that the axioms themselves are 'true'. We merely accept them on a kind of provisional basis. No doubt we have *some* kind of intuitive understanding that leads us to choose some axioms rather than others. But the mathematics itself — the thing we feel certain about — doesn't begin until *after* these axioms are in place. And once they are, a proof can stand perfectly well on its own logical feet without the support of this intuitive understanding. On this view, the actual substance of the mathematics — the logical edifice built upon the axioms — is entirely decoupled from whatever informal thoughts might have given rise to those axioms in the first place. In short, the practice of mathematics, and the criteria for its correctness, are clearly divorced from considerations of meaning.

Formalization: the benefits

So much for how the ideas of formalism came about. But what impact has all this had on mathematics today? In fact, the influence of the formalist

attitude has been profound, and I think all mathematicians, whatever their philosophical persuasion, would agree about some of the significant benefits it has brought.

For one thing, the formalization of mathematics has helped to provide absolutely clear ground rules for mathematics which can be accepted by large numbers of people. For instance, a large sector of the mathematical community today professes to sign up to a formal framework known as *Zermelo-Fraenkel set theory* (ZF for short), which is basically just a dialect of our glyph language with some vocabulary and axioms for set theory. (We'll be looking at ZF in detail in Chapters 9 and 10.) This means that, amongst those who subscribe to this framework, there's an absolutely clear standard for what counts as a proof. If you can say enough to show how your mathematical argument could, in principle, be translated into ZF, it will be accepted by the community. That's not to say that you actually have to *do* your proof in ZF (heaven forbid!) — the point is just that, if challenged, you should be able to show how this could be done for any given part of your proof. So if you and I both sign up to this framework, then even though we might have radically different ideas about whether mathematical objects really 'exist' or what they 'are', we'll be able to collaborate happily on mathematical problems without falling out, because the formal system provides a clear standard of proof and draws a sharp boundary between the mathematics and the philosophy.

Actually, not everyone does subscribe to ZF (more about this later), but other formal systems might play a similar role for other mathematical 'sub-communities'. What's more, such systems can be useful when it comes to communication between different sub-communities, because if we know *in which* formal systems a given proof can be conducted, then we all know where we stand.

The clean, hard edges that formalism has to offer are surely a big part of its appeal. Mathematicians by nature love precision, sharpness and clarity, and feel that their subject possesses these qualities *par excellence* — in marked contrast, perhaps, to the general fogginess, flabbiness and above all endlessness that many associate with philosophical discussions *about* mathematics. (Compare the question 'What is 5 + 3?' with 'What *is* the number 5 anyway?') So a view of mathematics that builds in a guarantee that we'll never get bogged down in interminable philosophical wrangling is bound to have a certain attraction.

Another benefit of a rigorously axiomatic approach is that the same mathematical theory can often be found to apply to many different situations. Let's suppose we've developed a theory of geometry based on certain axioms, using words like 'point' and 'line', and drawing inspiration from what we ordinarily understand by these words. Then one day we notice that exactly the same axioms apply if we invest the words 'point' and 'line' with some completely different meaning — for instance, we might find that tables and

beer mugs enjoy exactly the same basic properties. Now, because all our theorems have been deduced from the axioms *by logic alone*, we can conclude straightaway that all these theorems will be true under both interpretations. In other words, we get two theorems for the price of one, because the same proof applies in both cases.

Of course, that's a fanciful example: no one really expects that tables and beer mugs will satisfy the axioms of geometry in any useful sense. But here's a more realistic scenario. There is an important subject called *projective geometry*, with the remarkable property that if you interchange the words 'point' and 'line' throughout the axioms, you get a precisely equivalent set of axioms. In other words, it makes just as much sense to understand 'point' to mean what we'd normally call a line, and 'line' to mean what we normally call a point. This means that for every theorem you prove in projective geometry, you can get another theorem for free simply by swapping the terms 'point' and 'line' everywhere in sight.

There are many much more sophisticated examples of this principle of multiple application, and there's no doubt that the axiomatic method has enormously enriched our perspective on mathematics by allowing us to pick out recurring patterns and similarities, often relating what were previously thought to be totally unrelated areas of the subject.

So formalism as a *methodology* has a lot to offer. But as I hinted earlier, it's also possible to consider formalism as an actual *philosophy* of mathematics — one which denies or downplays the concept of meaning in some way. So let's now turn our attention to this idea. We shall consider three slightly different brands of formalist philosophy in turn.

Game formalism

The first, and purest, brand is one I'll call *game formalism*. This view at least has the merit of being easy to grasp: it claims that mathematics is ultimately *nothing more than* a game played according to certain simple rules with meaningless marks on paper.[16] So to ask what a mathematical statement 'means' is as pointless as asking what a position in chess 'means'. Here's how the mathematician Johannes Thomae put it in 1898:

> The formal conception of the numbers … does not ask what the numbers are or must be; rather it asks what one requires of the numbers in arithmetic. For the formalist conception, arithmetic is a game with signs that are well referred to as empty, by which one means to say that they (in the calculating game) have no content given to them beside that which comes from the role in the rules of combination … The formal standpoint allows us to throw off all metaphysical difficulties. That's the gain it offers us.[17]

A minor variation on this view (espoused by the logician Haskell Curry), is that mathematical theorems *do* say something meaningful — but what they say is *just* that certain strings of symbols can be obtained from certain other strings of symbols by such-and-such a system of rules. There's perhaps not much to choose between these two variants.

This might strike you as a very extreme line to take. Does any mathematician, deep down, *really* believe that the problems that engage him or her so passionately are nothing more than symbol-pushing puzzles? I used to think that no one really believed this, and that anyone who said they did was being disingenuous. Nowadays, I would say that if someone claims they find game formalism to be a completely adequate philosophy for what they personally do, then it's not for me to gainsay them. In one sense, the game formalist is at an unfair advantage: mathematics certainly does *involve* a fair amount of juggling with symbols, so it's possible to make game formalism true by definition, just by drawing the boundary of what we call 'mathematics' so as to include *only* this particular activity. The issue, of course, is that not everyone would agree that this is the most interesting or useful place to draw this boundary.

Game formalism is perhaps at its most plausible when it comes to highly rarefied branches of mathematics far removed from conceivable applications (we shall get a glimpse of some such ethereal realms in Chapters 9 and 10). For any mathematics that plays an important role in science, though, a natural question arises: if maths is just a game of symbols, why is it that it fits so well with our observations, and is so good at helping us make predictions?

At this point, questions of one's philosophy of science come into play. For instance, a game formalist might plump for an *instrumentalist* view of both scientific theories and the mathematical theories that support them. According to such a view, all we should ever claim about these theories is that they are *useful*, and that our belief in their usefulness is supported by experience. An instrumentalist would stop short of the idea that the reason why they're so useful is that they are, in some sense, *true* — or are at least good approximations to the 'truth'. Just as our scientific theories are selected according to this criterion of usefulness, so (one might say) our choice of mathematical theories — in order words, which versions of the formula game we choose to play — is guided by the same criterion, and this is all the explanation that's needed.

So the idea here is that the maths that physicists or biologists use is indeed useful, but carries no actual meaning or truth in itself. I will confess that this view doesn't sit at all easily with me as a mathematician, though I can see (just about) how someone might hold it. However, it seems to me that pure game formalism comes into very deep trouble if we bring it right down to the level of basic number theory. For here, at least, certain mathematical

statements seem to be clearly 'meaningful' in the sense that they make hard, testable predictions even within the realm of mathematics itself. Let me explain this.

Take a statement of Arithmetic like following, which expresses the so-called *distributive* law for × and + from school algebra:

$$\forall x \ \forall y \ \forall z \quad x \times (y + z) = (x \times y) + (x \times z)$$

Starting from suitable axioms, we might prove this theorem by pure symbol-juggling. But once we've done so, we're in possession of a very definite and concrete piece of knowledge: namely, that if we choose any three *specific* Whole numbers we like — say x = 23, y = 42, z = 91 — and then evaluate the two sides of the equation using the methods we learned at school, we'll get the same answer:

$$23 \times (42 + 91) = 23 \times 133 = \mathbf{3059}$$
$$(23 \times 42) + (23 \times 91) = 966 + 2093 = \mathbf{3059}$$

It doesn't matter (for our present purposes) whether the symbols 3, 6, 7, × or + themselves mean anything — for all we currently care, they could be just marks on paper that we all spent part of our childhood learning to manipulate. The important point is that the theorem as a whole is making a definite prediction about what we'll see happen if we perform certain kinds of calculations — *even if these calculations are themselves just meaningless symbol-juggling*. So in this sense, the theorem surely has some genuine content or meaning.

A similar, but less trivial, example is given by Fermat's Little Theorem, which, as we saw in Chapter 2, makes definite (and in this case non-obvious) predictions about the results of calculations: you can take any prime number $p > 2$, calculate 2^p and divide by p — and lo and behold, the remainder will be 2! Would we be impressed by a game formalist whose reaction to this was: "I'm not interested; that's not part of what I'm calling mathematics"? It seems to me that a philosophy that doesn't recognize at least this level of 'meaning' fails to do justice to mathematics as she really is.

In fact, I think very few working mathematicians or philosophers today would, in the light of such considerations, seriously espouse pure game formalism as an adequate philosophy for the *whole* of mathematics. Indeed, many formalists would be eager to distance themselves from this extreme position — either by clarifying which parts of mathematics they were game formalists about, or by aligning themselves with some other kind of formalism. So let's now see what other options are available.

If-thenism

Our second brand of formalism takes seriously the ideas that the axioms we choose to start from are 'provisional', and also that there's no fixed meaning to the symbols they involve.

Suppose, for instance, we're working with some dialect of glyph language, and with some chosen set of axioms **X**, from which we've proved some theorem ⊙ P. We've seen that our glyph language can be 'interpreted' in a multitude of ways, corresponding to the balloons in our story. In some of these balloons, the axioms of **X** will happen to be true; in others, they won't. So even to ask whether they were 'true', we'd need to specify which balloon we had in mind.

Now, the idea of if-thenism is that we don't commit ourselves on whether the axioms *are* true, or even on which balloon we're talking about. Rather, we understand the content of our theorem ⊙ P to be: in *any* balloon in which the axioms of **X** happen to be true, P will also be true. This approach is 'formalistic' in the sense that it's not dependent on understanding what the symbols in **X** and P refer to — we get round the need for this by considering *all* possible interpretations of these symbols.

There are certainly situations in mathematics where this is clearly a good approach to take. Indeed, in many abstract branches of mathematics, we very deliberately choose systems of 'axioms' that lend themselves to multiple interpretations. For instance, there's a branch of mathematics whose purpose is to study structures called *groups*, which come in many shapes and sizes. Much of this theory can be built up from the so-called 'axioms for a group', and every individual group serves as a *model* for these axioms: a little world within which they are true. So in this case, the axioms just *define* what a group is, and an if-thenist reading will make our theorems say exactly what we intend: in *any* model satisfying the axioms for a group (or more succinctly, in any group), such-and-such will hold.

An if-thenist view is also eminently sensible in the case of geometry. As we saw in Chapter 1, Euclid would have viewed his axioms as true in some absolute sense; but nowadays, with our awareness of elliptic and hyperbolic geometry, we would take a more relative view. Here again, if-thenism does just what we want: a theorem of Euclidean geometry will be understood to be 'true whenever the Euclidean axioms are true' — that is, in any space that satisfies these axioms. Likewise, in higher set theory, where (as we'll see in Chapter 10) the question of 'which axioms are true' becomes highly problematic, an if-thenist reading of theorems has much to commend it.

In other situations, however, the if-thenist reading might seem overly coy. Take statements of Arithmetic, for example. We haven't yet looked at what a suitable set of axioms for Whole numbers might be (this will come soon), but there's a strong intuition that we understand perfectly clearly what Whole

numbers are, and we can see that the axioms say true things about them. Suppose, then, that someone proves Fermat's Little Theorem from these axioms. Isn't it needlessly cautious to read this as saying '*If* the axioms are true in some model, *then* the theorem holds in that model' — as though the premise were something we didn't want to commit to? After all, as we've just seen, Fermat's theorem makes definite testable predictions: do we really feel that the reliability of these predictions is conditional on some assumptions whose truth we're unsure about, and which will indeed be false under other interpretations?

All of this highlights a strange kind of equivocation in the attitude of many mathematicians which has often been noted. There's a standing joke that the typical mathematician is a platonist on weekdays and a formalist on Sundays.[18] In other words, mathematicians instinctively *feel* that they are engaged in uncovering some kind of objective, timeless truth, and they certainly go about their work as if that were the case. But if challenged on whether they really believe what this seems to imply — namely that there is some eternal non-physical realm of existence to which our minds somehow have access — they tend to back off and deny that they really believe any such thing. In the words of the French mathematician Jean Dieudonné:

> On foundations we believe in the reality of mathematics, but of course when philosophers attack us with their paradoxes we rush to hide behind formalism and say, "Mathematics is just a combination of meaningless symbols" … Finally we are left in peace to go back to our mathematics and do it as we have always done, with the feeling each mathematician has that he is working with something real. This sensation is probably an illusion, but it is very convenient.

Or as Paul Cohen put it, 'the Realist [*i.e.*, roughly, platonist] position is probably the one which most mathematicians would prefer to take.' But if the difficulties with this view upset him, 'he will rush to the shelter of Formalism, while his normal position will be somewhere between the two, trying to enjoy the best of two worlds.'

Another difficulty with if-thenism is that the interpretation of mathematics it offers us seems at best incomplete. On the face of it, at least, the if-thenist reading takes a relatively simple statement, say Fermat's Little Theorem, and tries to explain its content via some much more complicated statement: 'In any model (structure, possible world, whatever) in which the axioms of Whole numbers hold, Fermat's Little Theorem also holds'. But this itself seems to be a mathematical statement, so our philosophy ought to include an explanation of what status *this* has. If this latter statement is itself meaningless, what progress have we made? Or if it means something, haven't we abandoned the essence of our formalist creed? And if it does have a meaning, what sort of

meaning does it have? Where do all these 'models' live? What on earth is a 'possible world'? We seem to have come round in a circle.

In summary, then, the if-thenist point of view is clearly a very helpful one to adopt in many circumstances, but once again, it doesn't seem by itself to provide a complete and satisfactory account of the whole of mathematics.

Hilbert's programme

By far the most interesting attempts to articulate a coherent and workable version of formalism, in my view, are those that stem from an idea introduced by David Hilbert and his followers in the 1920s. The wider story of Hilbert's programme — how it came about, what happened to it, and where later logicians have gone with the idea — is quite far-reaching, and permeates much of this book. For now, I want to focus on what the idea was in its original form. But to do this, we shall need to open up a complex of ideas of which Anna and Emily (for reasons of plot structure) are at present quite unaware, although they will arrive at them in due course.

Hilbert's programme was a grand proposal which, if it worked, promised to meet all of the above-mentioned difficulties head-on, and to lay to rest once and for all any nagging doubts people felt about the nature and status of mathematics. The idea was simple, elegant, ingenious and highly plausible.

Unlike the pure game formalists, Hilbert accepted that at least *some* mathematical statements had 'meaning': namely, those whose content could be explained in hard, practical terms, without reference to some 'metaphysical' notion of truth. But many other mathematical statements — in particular, statements that smacked in any way of the notion of *infinity* — he treated as mere combinations of symbols lacking any genuine content. Thus, Hilbert drew a clear distinction between *real* or *contentual* mathematical statements (those that actually meant something) and *ideal* ones (those that didn't). I'll be more specific in a moment about which kinds of statements Hilbert accepted as 'real'; but for now, suffice it to say that he took a pretty tough line, according to which even 'There are infinitely many primes' was classified as an 'ideal' statement.

Now, if there were no interplay at all between these two classes of statements, there would be no problem: we could accept the one as genuinely contentful, and dismiss the other as mere idle recreation. But it's not as simple as that. In practice, many, if not most, proofs of real mathematical statements make use of ideal ones: in other words, ideal statements appear to be an essential *tool* for proving real ones. Just as Dirichlet's proof of a statement about Whole numbers involved a long detour through Complex number land, so most proofs of 'meaningful' statements make important use of 'non-meaningful' ones. A striking example would be Wiles's proof of Fermat's last

theorem — itself a real statement even by Hilbert's strict criterion, but with a proof that exploits the power of ideal mathematics to an exceptional degree.

But this raises an obvious question. In what sense can a proof *be* a proof if most of the statements that appear in it are meaningless? How is such a 'proof' going to convince anyone of anything? How can a meaningless statement ever 'imply' a meaningful one, or indeed anything at all?

One radical response here might be to deprive oneself altogether of the use of ideal statements, even for proving real ones. On this view, the only acceptable proofs would be those that consisted entirely of real, meaningful statements. But Hilbert felt that this would lead to a drastic mutilation of the entire subject which he regarded as wholly unacceptable. He was himself steeped in the mathematics of infinity, and was deeply attached the richness and magnificence of the conceptions that it gave rise to. Besides, there was the challenge of explaining just *how* all this supposedly meaningless mathematics turned out to be as useful as it was. The task Hilbert faced, then, was to explain how meaningless statements can *legitimately* be used in proofs of meaningful ones.

The central idea is very appealing. First, Hilbert proposes to *formalize* all the rules of reasoning we wish to allow to use — those for ideal statements as well as real ones — in a precise symbolic system like the one we've been looking at. Next, he proposes to prove a grand meta-theorem about this system: namely that it is at least *consistent*. For reasons we'll see in the next chapter, this will have the effect of establishing that any given 'real' mathematical statement that can be proved in this symbolic system will, in fact, be *true*. And once we've established this, we're free to carry on using ideal statements with a clear conscience, exploiting to the full the power and intellectual economy they offer us, but secure in the knowledge that they'll never lead us to any 'real' but false conclusions.

Hilbert drew a compelling analogy here with one of the examples of formalism we mentioned earlier: the use of Complex numbers. For the sake of the analogy, let's suppose that statements about Real numbers have some genuine meaning for us (so that the 'real' mathematical statements are the 'Real' ones, if you like). Then one day, we're introduced to the powerful black magic of Complex number theory. Suppose that as far as we can see, this is mere meaningless symbol-juggling. But on the other hand, we can see that the magic *is* powerful and can be used to solve a lot of problems about Real numbers that we're interested in. So we're anxious to discover if there's a way of *justifying* the use of Complex numbers, without necessarily believing that such numbers 'exist'. In this case, one can indeed prove quite easily that every Real number statement that can be proved using Complex numbers could (perhaps more long-windedly) be proved using just Real numbers alone. So once we know this, we need have no qualms about the use of Complex

number magic. In effect, Hilbert wanted to apply a similar strategy on a much larger scale to justify the use of all 'ideal' statements in mathematics.

There's one final point to explain. For Hilbert's strategy to be convincing, the proof of our grand meta-theorem must *itself* be carried out entirely within the *real* part of mathematics — the part we regard as meaningful and unproblematic. That seems plausible enough, because it turns out that the meta-theorem we want — saying that our formal system is consistent — can indeed be cast as a real statement. If we can pull this off, then in effect we will have bootstrapped ourselves up from real mathematics to ideal mathematics, fully justifying the use of the latter without needing to suppose it 'means' anything.

I hope that this outline has managed to convey an impression that this basic strategy is, on the face of it, both highly attractive and extremely plausible. It somehow has the feel of something that's *bound* to work, and even today, a paper such as Hilbert's *On the infinite* (1925) can come across as quite persuasive. And if it did work, it would certainly give about as satisfactory a foundation for mathematics as one could wish for. By the end of the 1920s, Hilbert and his followers staunchly believed that the goal was in sight, and that it was just a matter of filling in the details:

> With this new way of providing a foundation for mathematics … I should like to eliminate once and for all the questions regarding the foundations of mathematics … I believe that I can attain this goal completely with my proof theory, even if a great deal of work must still be done … Already I should like to assert what the final outcome will be: mathematics is a presuppositionless science.[19]

You might wonder how all this fits in with the other big idea of Hilbert's which we first met at the end of Chapter 2: his famous decision problem (the *Entscheidungsproblem*). In fact, both ideas formed part of the same grand scheme, but the idea of the decision problem came in around 1928, a bit later than the ideas we've just been discussing. As we saw in Chapter 4, the work of Church and Turing demolished any hopes of solving the decision problem. Whether the other half of Hilbert's project suffered a similar fate, we shall learn in the next chapter.

A spectrum of formalisms

But we're in danger of getting distracted here. The story of Hilbert's programme is fascinating for many reasons, but the aspect of it that I want to focus on in this chapter is the particular flavour of formalism that it proposes.

Even from this point of view, there are at least two reasons for being interested in Hilbert's programme. First, it introduces us to a very general

idea: that of drawing a line between some class of 'real' statements, which we consider meaningful, and a class of 'ideal' ones, which we initially consider as meaningless — and then looking for some sort of justification for being allowed to use ideal statements in proofs of real ones. This basic strategy is a very adaptable one: we don't have to draw the dividing line exactly where Hilbert did, nor do we have to seek exactly the same kind of 'justification' for ideal statements that he sought. Indeed, we can imagine a whole spectrum of varieties of formalism arising from different ways of making these choices.

But another reason for being interested in Hilbert's version of formalism is that the particular place where he chose to draw the dividing line is about as low down as one can sensibly draw it. In other words, the class of statements he accepted as 'real' was about as small as one can get away with, without falling into the sorts of problems we met in connection with pure game formalism. Correspondingly, his class of 'ideal' statements was taken to be as large as possible. Thus, Hilbert's stance represents a kind of extremal point on the spectrum of how formalistic it's possible to be without doing violence to the very essence of mathematics. That alone is enough to make Hilbert's standpoint interesting. We shall therefore spend some time looking at where Hilbert drew the real/ideal boundary, and then trying to think ourselves into the mindset of someone who adopts this conception of mathematics.

We mentioned earlier that all statements that made reference (either explicitly or implicitly) to the idea of *infinity* were classified by Hilbert as 'ideal' statements. In other words, the only statements he accepted as meaningful were those that referred only to finite entities and concepts — a position Hilbert himself labelled as *finitism*. Thus, the words 'finitism' and 'formalism' both describe aspects of Hilbert's position: typically, it's natural to refer to his attitude as 'finitist' when we have in mind those statements he did accept as meaningful, and as 'formalist' when we have in mind those he didn't. That said, even the 'meaning' he attached to finitary statements was quite formalistic in character, as we'll see.

One might perhaps wonder whether Hilbert, an exceptionally great mathematician who made deep and fundamental contributions to infinitary mathematics, can *really* have believed in his heart that it was ultimately only finitary statements that were truly meaning (or 'contentual' as he put it). Actually, there's quite a bit more to say about the historical context within which Hilbert was arguing this, and we'll come to this in Chapter 11. For the time being, however, we shall simply note that Hilbert's papers from the 1920s leave no room for doubt that this *was* his stated position, so we shall take him at his word, and I shall often refer to the finitist position as 'Hilbert's view'.

Our first task, then, is to understand exactly which kinds of statements Hilbert accepted as finitary and therefore meaningful. Actually, there are two different categories of statements that we'll need to consider: I shall call them

Checkable and *Predictive* statements. These will play key roles not just in our account of Hilbert's programme, but throughout the rest of the book.

Checkable statements

Roughly speaking, *Checkable* statements are concrete statements whose truth or falsity we can checked by some direct, routine calculation which we know can be completed within a finite time. Some simple examples would be:

$$0 = 1 \qquad 2 + 8 < 3 \times 4 \qquad (3 \times 3) + (4 \times 4) = 5 \times 5$$

Some of these statements are true and some are false, and we all know how to determine which are which. And as we noted earlier, this is something we can agree on even if we disagree about whether the symbol '4' actually refers to anything. In other words, we can give truth-values to these statements via a purely symbolic calculation procedure, without entering into any metaphysical debate about what numbers 'are'. The idea is that even if we refuse to associate any meaning with the symbols, we can simply *define* a statement to be true if our conventionally agreed calculation procedure yields 'true'.

We can extend our language a bit to allow other expressions such as 2 ^ 3 (for 2 to the power of 3) and 5! (for 5 factorial), and the same principle will obviously still apply, although some of the calculation steps will start to get more length at this point. In fact, Hilbert gave a quite general description of a class of other operations you could safely add whilst still maintaining this property of 'checkability by direct calculation within a finite time'.[20] For simplicity, though, I'll stick with the numerical operations I've already mentioned (+, ×, ^ and !), which are enough to illustrate the general idea.

There are also other glyph statements that we can reasonably classify as Checkable. It's clear that if we take some Checkable statements and combine them using the propositional glyph symbols ¬, &, ∨, ⇒, then the resulting statements will still be finitely checkable. For example:

$$\neg\, 1 = 5 \qquad 1 < 2 \;\&\; 2 + 7 = 3 \times 3 \qquad 1 = 5 \;\vee\; 45 > 23 + 1$$

Indeed, we can check these by checking the truth-values of the basic statements they contain (such as 1 = 5), then using the truth tables from Chapter 3 to work out the truth-value of the whole formula.

What sorts of statements *aren't* Checkable? A simple example would be

$$\forall x \quad x < x + 1$$

Although this is very obviously true, it can't be checked just by *direct calculation* in the way we have in mind here, because that would involve checking the formula x < x + 1 for infinitely many values of x:

$$0 < 0 + 1\,, \qquad 1 < 1 + 1\,, \qquad 2 < 2 + 1\,, \qquad 3 < 3 + 1\,, \qquad \ldots$$

Each of these statements individually is Checkable, but the fact that they're *all* true isn't something we can directly check just by finite, routine calculation. Indeed, no statement that something holds in infinitely many cases will count as Checkable in our present sense.

However, there are certain rather special kinds of statements involving '∀' which we *can* admit as Checkable. At this point, I'd like to propose a small extension to our glyph language, in order to allow formulae like

$$\forall x < 10\,.\;\; x < x + 1$$

The new thing here is the sequence '< 10 .', and the intended meaning (although as formalists we're not really supposed to care about this) is 'for every Whole number x less than 10, x is less than x + 1'. Actually, you could see this as simply a convenient abbreviation for the following formula in standard glyph language:

$$\forall x \quad x < 10 \;\Rightarrow\; x < x + 1$$

Either way, the point is that adding the condition 'x < 10' makes the statement routinely checkable, because we only need to check the relevant property for finitely many values of x, namely 0, 1, …, 9. So even if we don't agree on whether '∀' means anything, we can at least agree on the calculation procedure that we'd use to check the above statement, then simply *define* the statement's truth-value to be the result of this calculation.

In the same way, we can allow formulae involving ∃ such as

$$\exists x < 30\,.\;\; (x \,\hat{}\, 2) + x = 182$$

with the intended meaning: 'There exists a Whole number x less than 30 such that x squared plus x is 182.' (Little exercise: how would you rephrase this in standard glyph language?[21]) Once again, we can regard this as Checkable, because to determine whether it's true, it's enough to check the relevant property for the finitely many values 0, 1, ..., 29.

To summarize, a glyph formula can involve the symbols ∀ and ∃ and still be Checkable, as long as all the uses of these symbols are qualified by a 'less-than' condition that puts an upper limit on the numbers we need to check.

Broadly speaking, the spirit of Checkability remains the same: we can obtain a truth-value by a direct calculation which we know will be completed in a finite time, and what's more, we can say in advance how long (at worst) this calculation is liable to be.[22]

That basically completes our description of Checkable statements. The only other thing to say is that we'll also be considering Checkable formulae with free variables. For example, here's a Checkable formula that's constructed according to the means I've described, except that it involves the free variable x. Its (secret) intended meaning is 'x is a prime number'.

$$1 < x \;\&\; \neg\,(\exists y < x \,.\, \exists z < x \,.\, y \times z = x)$$

This is no longer a statement whose truth-value we can determine outright. Rather, it is a *property* whose truth or falsity we can check for any given value of x we care to supply. For instance, we could substitute '23' for 'x' throughout this formula, and this will give us an ordinary Checkable statement whose truth value we could routinely calculate. This shows that 'x is a prime number' is a Checkable property of x in our sense.

It's also worth noting that Checkable statements can easily be checked by a suitable tube machine, though by no means every property decidable by a tube machine is Checkable. So we should think of Checkable properties as a particularly simple class of decidable properties: roughly speaking, what's distinctive about them is that they can be checked by a finite calculation, and moreover the finiteness of this calculation doesn't itself require any special proof, since it's automatic from the way the properties are constructed.

Perhaps the hardest aspect of all this is simply learning to think ourselves out of our usual habit of associating 'meanings' to formulae, and viewing of them just as symbol sequences that we know how to manipulate mechanically. In the last few pages, just to help you along, I've occasionally indicated the intended 'meanings' of some of our formulae, but the Formalist Police wouldn't really approve of this. We need to get used to the idea that the truth-value of a formula is just *defined by convention* to be the result of the blind symbol-pushing calculation, nothing more (although if you do secretly believe in some other sort of meaning, you'll concede that in the case of Checkable statements it agrees with these symbolic calculation results). In other words, it's only the existence of this 'verification procedure' for Checkable statements that *makes* them true or false. It is in this limited sense, and this alone, that Checkable statements can be said to be 'meaningful'.

Predictive statements

On any reasonable view of mathematics, Checkable statements can't be the whole story. As Emily learned in Chapter 1, mathematics isn't just about doing concrete calculations with specific numbers: the whole beauty of the subject is that it gives us a way to prove (for instance) that some property holds for *all* Whole numbers. In order to account for this, we therefore need to admit a second category of statements which even our hard-headed formalist ideology can admit as being in some way meaningful.

A good example of a statement about all Whole numbers is the distributivity law that we met a few pages back:

$$\forall x \ \forall y \ \forall z \quad x \times (y + z) = (x \times y) + (x \times z)$$

This isn't a *Checkable* statement, because it's saying something about *all* possible numbers x, y, z: we haven't put any upper limit on their size in this case. So it doesn't enjoy quite the same status of being directly verifiable by a finite calculation. However, we'd better be able to give such statements *some* status within our framework, otherwise we won't be able to do any decent mathematics at all.

It's perhaps worth clarifying what we're trying to avoid here. We *don't* want the criterion for the truth of the above statement to be based on a sort of 'infinite extension' of what we'd do for Checkable statements. In other words, we don't want to hypothesize anything like an angelic being who can check the distributivity equation for the infinitely many possible values of x, y, z, and then report back to us on whether it was always true. That would be completely contrary to the hard-nosed attitude we're trying to adopt here, because no one has such an angel (or any equivalent computational means) at their beck and call. So the question is: can we regard the above statement as genuinely meaningful or contentful in some other way, without compromising on our finitist philosophy?

Hilbert himself, in his writings, seems to equivocate a bit on whether he accepts such formulae as 'contentual' in themselves, and it remains a bit unclear just how, from his über-finitistic standpoint, he intended to regard the apparent allusion to the infinite in the above formula ($\forall x$, etc.). The situation has been more carefully analysed by later scholars,[23] and the consensus seems to be that Hilbert needs these '$\forall$' formulae to be contentual, given the use he wants to make on them. I'd like to offer here a way of thinking about it which I personally find illuminating, and which I think can't be *too* far from what Hilbert had in mind.

The key idea is one that has come up a few times already: the above statement means something because it makes a genuine, testable *prediction*

about what will happen if you choose specific numbers for x, y, z and then check whether $x \times (y + z) = (x \times y) + (x \times z)$ (the latter is of course a Checkable formula). So the distributivity law has the character of a kind of bet: *whatever* particular numbers x, y, z you choose, I bet that this equation will be revealed by direct calculation to be true. Or if you think that the 'whatever' there still smacks a bit of the infinite, we can bring the whole thing down to a finite, though hypothetical, situation: we can imagine that you've already chosen some numbers x, y, z; and without knowing what your numbers are, I'm now betting that they satisfy $x \times (y + z) = (x \times y) + (x \times z)$. Even if we don't believe in any concept of infinity, we can see this bet has a genuine content, in that we clearly understand what it would mean for it to be proved wrong and under what circumstances I'd have to pay up — so I'm genuinely sticking my neck out by claiming that these circumstances won't arise.

What exactly is it about the form of the distributivity law which gives rise to a bet in this way? You can see that it consists simply of a Checkable formula (the equation) preceded by three '∀' prefixes, and we are reading it as saying 'you choose any x, y and z, and I bet that your numbers will satisfy the Checkable property $x \times (y + z) = (x \times y) + (x \times z)$'. Indeed, it's easy to see that any statement with this basic structure will enable us to make a similar bet, and thus will have predictive content. This leads us to make the following general definition:

> A *Predictive* statement is any statement that consists of a Checkable formula headed by a sequence of '∀' prefixes.

In this book, I'm using the term 'Predictive' for what are unhelpfully known in the technical literature as Π_1 *statements*, for reasons I won't bore you with. We saw earlier that there's a problem for a game formalist who wants to regard statements of this form as completely contentless: namely, they must mean something, because they enable us to make a bet. But what we're now proposing is: let the bet itself *be* the content of the statement. Thus, Predictive statements are not meaningful in quite the same way that Checkable ones are, but they carry their own kind of meaning nonetheless.

There's an analogy here with a well-known idea in the philosophy of science. According to the philosopher Karl Popper, a theory can only be truly classed as 'scientific' if it is in principle *falsifiable* — not, of course, in the sense that it's actually false, but in the sense that there's some imaginable set of observations which, were they to occur, would refute the theory. A scientific theory, in other words, ought to make at least some definite predictions about how things will turn out to be: a theory that could be fitted to any conceivable observations whatsoever was, for Popper, not 'scientific'.

I'm not wanting to discuss the merits or demerits of Popper's view of science here, but simply to point out the analogy with our Predictive statements. Think of the verification of Checkable statements as the analogue of 'experimental observations' in science. A Predictive statement would count as a good scientific hypothesis in Popper's sense, because it commits to a definite prediction about the results of such 'observations': a single contrary observation would show the statement to be false. On the other hand, just as with a scientific hypothesis, no number of observations *in support of* a Predictive statement can definitively show it to be true, though they might well increase our disposition to trust it.

Predictive statements: examples and non-examples

It's worth looking at a few more examples of Predictive statements to get a feel for the scope of the concept. A particularly good example is Fermat's Little Theorem, which we can express schematically as

$$\forall p \ \forall c \quad (\text{`p prime'} \Rightarrow \text{`p divides c } \wedge \text{ p} - \text{c'})$$

Once again, this makes a definite prediction about something good which we can observe happening for any numbers p and c that we choose. We've already seen that being a prime number is a Checkable property, and not surprisingly the same is true for 'p divides c ^ p – c'. (You might enjoy constructing an actual Checkable glyph formula that expresses this.) So if we've proved Fermat's Little Theorem in a way which we feel is not open to doubt, we're armed with some very concrete knowledge that we can bet our bottom dollar on. And on the basis of this, as we saw in Chapter 2, we can go ahead and write some software to implement the RSA cryptosystem, secure in the knowledge that encrypting a message and then decrypting it will always give us back the original message. This is a good example of how a bet made by a Predictive statement might play out in its applications.

Here are a few more statements from number theory (not all proved!) which are either Predictive as they stand, or can be made so with a bit of tweaking.

Fermat's last theorem: For all x, y, $z > 0$ and $n > 2$, it's *not* the case that $x^n + y^n = z^n$. This is a Predictive statement as it stands: what it predicts is just that you'll never find a (Checkable) counterexample to the theorem. This is a good example because it's so hard to prove: indeed, at present, no one knows for sure whether it's possible to prove it without going beyond the realm of Checkable and Predictive statements.

Goldbach's conjecture (still open): Every even number greater than 2 is a sum of two primes. The only thing to note here is that 'n is a sum of two primes' is Checkable, because the two primes in question would both have to be less than n, so there are only finitely many possibilities to check. Indeed, we can express Goldbach's conjecture in Predictive form like this:

$$\forall\, n \quad (\text{`n even'} \,\&\, 2 < n \;\Rightarrow$$
$$\exists p < n \,.\, \exists q < n \,.\, \text{`p prime'} \,\&\, \text{`q prime'} \,\&\, p + q = n)$$

Lagrange's theorem: Every Whole number is a sum of four perfect squares (we allow $0^2 = 0$ as a perfect square). Again, the trick is to notice that being a sum of four squares is a Checkable property: if we're given a number n and are looking for numbers a, b, c, d such that $a^2 + b^2 + c^2 + d^2 = n$, we only need to consider numbers less than or equal to n itself.

Riemann hypothesis (still open): I won't go into this here, but it's known that this famous conjecture is equivalent to a Predictive statement.

So all of these theorems or open problems could be recognized as meaningful by our hard-line finitist who accepts only Checkable and Predictive statements.

What would be an example of a *non-Predictive* statement? Such examples are in fact very easy to find: indeed, we mentioned earlier that even 'There are infinitely many prime numbers' is an example. A typical way of expressing this in glyph language might take the form:

$$\forall n \quad \exists p \quad (p > n \;\&\; \text{`p is prime'})$$

First of all, we can note that this doesn't have the right structure to be a Predictive statement: the formula coming after the '$\forall$n' isn't Checkable, because the '$\exists$p' here doesn't come equipped with a finite upper limit.

To dismiss such a fundamental statement as meaningless might seem quite extreme, perhaps even shocking, so let's make the effort to think ourselves into a mindset from which this makes sense. Recalling the analogy with Popper's idea of falsifiability, the idea is that the above statement can be seen as 'not saying anything' in the sense that there's no conceivable set of 'observations' (i.e. calculation results) that would either confirm or refute it. To see this, suppose you know two professors, P and Q, and suppose P tells you "There are infinitely many primes," and Q tells you "No, there are only finitely many primes". You then collect a lot of data on which numbers are and aren't prime, in the hope that this might tell you who is right. On the one hand, suppose that for as far as you look, the prime numbers seem to keep on

coming. Then P can say "Just what I'd expect," but Q can also say "That doesn't prove anything: there *is* a largest prime number, you just haven't got there yet!" On the other hand, suppose you found that after a certain large number there seemed to be no more primes. Then again, Q can say "Great — perhaps you've now found where the prime numbers stop!" But P can also say "That's perfectly consistent with my view: there *are* more primes beyond that, but you haven't reached them yet." And it's clear that *whatever* data you collected — however far you looked — both P and Q could continue to maintain these positions. In other words, both 'There are infinitely many primes', and its contrary are compatible with any finite set of observations whatsoever. So if our philosophy is that Checkable statements are the only hard currency we accept — the only grounds for the truth or falsity of anything — then the statement of Euclid's famous theorem, taken in isolation, doesn't say anything at all.[24]

What Hilbert had in mind, then, was to work with (in effect) a severely restricted version of our language of Arithmetic in which only Predictive statements were allowed. This system, with suitably adapted Rules of Thought and appropriate axioms, would be all that was permitted for the development of real, meaningful mathematics. We'll see in more detail in the next chapter what such a system might look like. But to conclude this chapter, let's consider briefly where this Hilbertian philosophy might take us.

Where now?

As we've seen, Hilbert presents us with an austere view on which the only ultimate basis for meaning in mathematics is finite, symbolic calculation, so that whilst Checkable and Predictive statements may be said to be meaningful in a certain sense, nothing else can.[25] I personally think this is a coherent philosophy with a certain attraction to it, albeit one that will appear insufferably restrictive to the majority of mathematicians. So the question now is: supposing we buy this philosophy, where can we go from there? What do we make of the vast edifice of mathematics we are rejecting as meaningless, and in particular, of its use in proving meaningful theorems?

Several options are available. We could decide to limit ourselves entirely to a diet of meaningful statements, both in theorems and in their proofs, and see how much mathematics we're able to build on this restricted base. (Just how much of a sacrifice this choice would entail will be investigated in Chapter 12.) Or we could accept that meaningful theorems with meaningless proofs had sufficient usefulness to be retained, but abandon (or pretend to abandon) the idea that such 'proofs' conferred any special kind of certainty. Or we could hope, as Hilbert did, to cut our losses by showing that the use of (at least some) meaningless statements in a formalistic way was somehow justifiable

after all. We'll be looking both at Hilbert's original attempt to do this and — later on — at more recent efforts along broadly similar lines.

But we've jumped ahead somewhat. We've been talking about using formal systems to prove mathematical theorems. But so far, we've only seen one half of such a system: the part concerned with logical deduction. To complete the system, as we've noted several times, we also need some *axioms* — some basic facts about Whole numbers (or whatever else we're studying) which our proofs can start from. Identifying a good set of axioms for the theory of Whole numbers will form our first piece of business in the next chapter.

[1] More precisely, **A** ⊙ P should be read as saying: 'For every assignment of values to the free variables involved, if the formulae of **A** hold relative to this assignment, then P also holds relative to the same assignment.' For instance, in the case of our example, the free variables are x and y. So our theorem will say: 'For any values of x and y, if $4 \leq x$ and $5 < y$ then $9 < x + y$'.

In logic textbooks and papers, the 'turnstile' symbol ⊢ is usually used in place of our medallion symbol, and Greek capital letters such as Γ are often used in place of **A** as metavariables for sets (or lists) of assumptions. So a typical theorem (or *judgement* as it is sometimes called) would have the form Γ ⊢ P.

[2] The precise meaning of 'x not free in **A**' is that there are no *free-floating* occurrences of x within A, in the sense we explained in Chapter 3. For example, if one of the formulae in **A** is $x < y$, then **A** contains a free occurrence of x, and we'll be forbidden from applying the ∀ Intro rule. On the other hand, if the only appearance of x in **A** is within a formula $(\exists x\ \ x < y)$, that won't matter, because the occurrence of x here is *bound*: that is, the 'x' in $x < y$ is taken to refer to the 'x' introduced by the ∃x, which we think of as 'a different x' from the one we're wanting to deal with. If all occurrences of x within **A** are bound in this way — that is, they all occur within some subformula (∀x …) or (∃x …) — then the ∀ Intro rule is still permitted.

[3] To clarify how P[x→T] is defined, there are two technical points to make. First, we want to replace just the *free* occurrences of x within P by T (see the previous endnote). For example, if P is the formula $x < y$ & $(\exists x\ \ z < x)$, then the first occurrence of x is free but the others aren't (they refer to 'a different x'), so P[x→T] will have the form $T < y$ & $(\exists x\ \ z < x)$.

The second issue is rather tiresome. Suppose P is $(\exists y\ \ x < y)$ and T is $y + 5$. Since the x within P is free (it's not bound by ∃x or ∀x), it seems that P[x→T] should be $(\exists y\ \ y + 5 < y)$. But what has happened here? The 'y' in $y + 5$ has been 'captured', and now refers to the y introduced by the ∃y, which is really a different y.

The slightly messy solution that's usually adopted is that when we perform a substitution, it may be necessary to *rename bound variables to avoid capture*. The idea is that the actual choice of which variables we use is unimportant, and there's clearly no essential difference between writing $(\exists y\ \ x < y)$ and $(\exists y'\ \ x < y')$. So the idea is that just before we do the substitution, we can replace the first of these by the second, so that the substitution is now 'safe',

resulting in (∃y′ y + 5 < y′). To do this, we need some method for choosing a suitable fresh variable to use (in this case y′), but it's easy enough to dream one up.

You might feel that this is all rather ugly and artificial, and you'd be right. Really, this ugliness comes about because, in the general interests of human readability, we're working with a *textual* representation of formulae as sequences of symbols. If one chooses to represent formulae in a less textual way, say as graph structures of some kind, then this ugliness disappears and the rules of logic appear as beautiful as they ought to be. But if we did this, books and papers on logic would be even more unreadable than they are.

[4] The best examples to date are the formalization by George Gonthier of the famous *four-colour theorem* for planar graphs, and Bob Hale's proof of Kepler's *sphere-packing conjecture*.

[5] Besides the point discussed here, two other aspects of Gödel's original work deserve mention. First, his theorem applied not just to our Whole number language but to *any* dialect of glyph language. Second, he also noted that a similar result would hold if the relation '=' were treated as part of logic, with some suitable axioms added to the system. (We will introduce the axioms for '=' in Chapter 6.)

[6] From G. Parkinson, *Leibniz — Logical Papers*, Oxford University Press (1966), page 105; quoted in M. Davis, *Engines of Logic*.

[7] In G. Boole, *An Investigation of the Laws of Thought* (1854), Chapter 1.

[8] From D. Hilbert, *The foundations of mathematics* (1927), reproduced in Van Heijenoort (1967), page 475.

[9] A fascinating collection of articles representing a wide variety of viewpoints (aimed at a philosophically literate reader) appears in P. Boghossian and C. Peacocke (eds.), *New Essays on the A Priori*, Oxford University Press (2000).

[10] As late as 1770, Leonhard Euler wrote: "… it is clear that the square roots of negative numbers cannot be included among the possible numbers. Consequently we must say that these are impossible numbers. And this circumstance leads us to the concept of such numbers, which by their nature are impossible, and ordinarily are called imaginary or fancied numbers, because they exist only in the imagination."

[11] This idea was made fully explicit by William Rowan Hamilton in 1837, though it seems to have been implicitly understood by several others some time earlier.

[12] From a letter from Leibniz to Fr. Des Bosses in 1707; quoted in M. Detlefsen, *Formalism*, in Shapiro (2005).

[13] According to the traditional stereotype, pure mathematicians are unhappy with any result that is not based on an impeccable logical foundation, whereas applied mathematicians are quite happy to use methods that appear to give them useful and meaningful results, even if they don't have a watertight logical justification for them.

Interestingly, some prominent mathematicians have voiced concerns that today's 'purist' preoccupation with logical rigour could have a stifling effect. Sir Michael Atiyah writes: 'But if mathematics is to rejuvenate itself and break exciting new ground it will have to allow for the exploration of new ideas and techniques which, in their creative phase, are likely to be as dubious as in some of the great eras of the past. Perhaps we now have high standards of proof to aim at but, in the early stages of new developments, we must be prepared to act in more buccaneering style.' (In M. Atiyah et al., *Responses to "Theoretical Mathematics" by Jaffe and Quinn*, Bulletin of the American Mathematical Society 30(2) (1994), p.178.)

[14] Here is a textbook example of such sheer nonsense, taken from the theory of infinite series, a subject closely related to the calculus. There's a famous and beautiful result that the value of

$$1 - 1/3 + 1/5 - 1/7 + 1/9 - 1/11 + 1/13 - \ \ldots$$

is exactly $\pi/4$, which is around 0.785395. You'll notice that this series consists of an alternation of positive terms (1, +1/5, +1/9, +1/13, …) with negative ones (–1/3, –1/7, –1/11, …). But now let's shuffle the series around so that exactly the same terms appear but in a different order. (Surely the order in which we add things together doesn't matter? We all learn that at school, after all.) We'll reorganize the series so that we take two positive terms for each negative one, but every term in the original series still appears somewhere in the new one:

$$1 + 1/5 - 1/3 + 1/9 + 1/13 - 1/7 + 1/17 + 1/21 - 1/11 + \ \ldots$$

Now the value of the series is suddenly different: it's around 0.958685. So it would seem that just by applying some familiar rules in this unfamiliar situation, we've shown that 0.785395… = 0.958685…. (Even more weirdly, you can actually cause the value of this series to be *any Real number you like* just by reshuffling the terms in a suitable way!) This illustrates the need for some clearer ground-rules on what kinds of manipulations are allowed.

[15] From H. Hahn, *Empiricism, Logic and Mathematics*, reprinted by D. Reidel (1980). Quoted in Detlefsen's article in Shapiro (2005).

[16] I've borrowed my wording here from a quotation often falsely attributed to Hilbert. It first appeared in Bell (1952) and has since

achieved wide currency. Whilst this quotation offers a good description of what game formalism is, it doesn't chime in at all with Hilbert's declared views, which (as we shall shortly see) were considerably more subtle than those of the game formalist. This spurious quotation has been responsible for great mischief.

[17] From J. Thomae, *Elementare Theorie der analytischen Functionen einer complexen Veränderlichen*, 2nd ed., L. Nebert, Halle (1898); quoted in Detlefsen's article in Shapiro (2005).

It might be considered misleading to cite Thomae as an advocate of the kind of game formalism we have in mind here, since he was speaking only of 'arithmetic' and didn't have the general concept of a formal system. On the other hand, it's hard to find clear quotations in support of pure game formalism from prominent mathematicians of *any* period. Another candidate might be John von Neumann: 'We must regard classical mathematics as a combinatorial game played with the primitive symbols' (*Theory of Self-Reproducing Automata* (1966), pages 50-51).

[18] See Davis and Hersh (1981), Chapter 7, to which I am also indebted for the quotations from Dieudonné and Cohen that follow.

[19] From D. Hilbert, *The foundations of mathematics* (1927), reproduced in Van Heijenoort (1967), page 479.

[20] Specifically, the class of operations Hilbert introduced was the class of *primitive recursive* functions. These can be defined as the functions you can construct from a few very simple ones via the mechanisms of *composition* and *primitive recursion*. I won't define these here, but you can get a feel for what primitive recursion is by seeing how it can be used to introduce the new operations '^' and '!', assuming we already have '+' and '×'. We can define these by just two equations each:

$$x \wedge 0 = 1\,, \qquad x \wedge (y + 1) = y \times (x \wedge y)$$
$$0! = 1\,, \qquad (y + 1)! = (y + 1) \times (y!)$$

[21] The correct formula is $\exists x\,(x < 30\ \&\ (x \wedge 2) + x = 182)$. A not uncommon mistake is to use '$\Rightarrow$' instead of '&' here. This doesn't do what we want, because the formula $x < 30 \Rightarrow (x \wedge 2) + x = 182$ actually comes out *true* for any value of x bigger than 20, which isn't what we intend. (Recall that 'false $\Rightarrow$ false' is 'true' according to the truth table for $\Rightarrow$ that we saw in Chapter 3.)

[22] This is a reasonable intuition to have for the examples we shall encounter, for which the maximum length of the calculation can be given by a fairly simple formula. But in general, there's a bit more to this than meets the eye. For very complicated Checkable

statements, telling how long the calculation will be might *itself* involve some significant calculation, and we might then worry about how long *this* is going to take. And working *that* out might itself involve a third calculation, whose length would be determined by a fourth calculation, and so on…. At the end of the day, there's nothing to worry about, because all these calculations are guaranteed to be finite — but saying their lengths are 'known in advance' rather glosses over something.

[23] See in particular the first two chapters of Tait (2005) for a careful explication of the finitist position. In endnote 4 to Chapter 2, Tait notes that universal statements must be considered contentual both because the consistency statement Hilbert is after is a universal statement, and because one premise of the induction rule (which had itself better be contentual) is of this kind.

Much as I value Tait's analysis, it is not so easy to present it in non-technical terms. I hope that the explication I offer here, though perhaps crude by comparison, will at least afford novice readers an initial impression of the spirit of Hilbert's finitism and of how it fits into our wider picture.

[24] Actually, in this particular case (as Hilbert himself pointed out), there's an interesting tweak we can make to bring Euclid's theorem into the fold of Predictive statements. If we delve into the *proof* of this theorem (the one Emily saw in Chapter 1), we can see that the method actually establishes that for any n, there's a prime number greater than n *but no greater than* n! + 1. So if we incorporate this extra information into the theorem, we get a statement that is indeed Predictive, because there's now an upper limit on p:

$$\forall n \;\; \exists p < n! + 2 \; . \;\; (p > n \;\; \& \;\; \text{`p is prime'})$$

This shows that at least in some cases, it may be possible to *adapt* some 'non-Predictive' piece of mathematics in order to shoe-horn it into the Predictive world. But this will typically require some real work, and it may not always be possible. So the point remains that on the philosophy we're considering here, we can't simply and uncritically continue to use non-Predictive statements as they stand without further justification. We shall explore this line of thought further in Chapter 12.

[25] You might wonder about the class of statements that are 'dual' to Predictive ones: those consisting of a Checkable formula headed by some ∃ prefixes, such as ∃p (1000000000 < p & 'p prime'). Such a statement would not satisfy the falsifiability criterion, but could of course be *verified* if we found a number p with the relevant property. It would be natural to see this statement as the 'negation'

of ∀p ¬ (1000000000 < p & ‘p prime’).

Hilbert explicitly ruled out such statements, declaring that there were statements whose negation wasn’t an allowable statement. One might argue that we should be allowed to assert ∃p (1000000000 < p & ‘p prime’) if we have an actual example of a suitable p (such as 1000000007). But it wouldn’t really extend the power of our framework to allow this, because ∃p (1000000000 < p & ‘p prime’) wouldn’t tell us anything that (1000000000 < 1000000007 & ‘1000000007 prime’) didn’t. In this sense, there’s an important asymmetry between Predictive statements and their supposed duals.